Attempt-at-the-hard-problem

Causal Access: A Decision-Theoretic Foundation for Representation, Selection, Report, and Agency in the Science of Consciousness

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Abstract. Consciousness science lacks a shared, formally stated account of what it means for a content to be accessed. Decoding studies show that information is present, not that anything uses it. Reports are treated as read-outs of access, yet reports have their own inputs and biases. Capacity limits are invoked without saying which resource is limited. This paper proposes a foundation for the functional part of the problem. Access is defined as causal use by independently specified readers, measured as a profile of decision value, intervention effect, and sensitivity. A workspace is distinguished from a mere relay by selection: with the input fixed, internal state or competing input determines which content enters shared use, at a cost to rivals, and in response to task value. Resource limits are modeled as constrained decision problems, and I prove that optimality claims have no content without an independently measured loss. Report, introspection, self-modeling, valence, unity, and thermodynamic cost are treated as separately measurable dimensions. I prove twenty propositions and three corollaries, most of them elementary, and use them to work through fifty problems from psychophysics, neurology, psychiatry, philosophy, and machine learning. Several give sharp consequences: log-error conservation under a saturated budget, exclusion thresholds that give the direction of perceptual load and extinction effects, conditions under which serial access beats sharing, and a covariance signature that shared entry must show, though other causes can show it too. Some of these meet evidence that already complicates them, which I report. I also show where the framework fails to explain. It does not derive feeling from function. Any step from organization to experience requires a bridge hypothesis, and I display the verdicts each candidate bridge gives. No empirical test of the framework has yet been run, and none of the worked accounts should be read as confirmed.

What this work achieves and what it explains

What it achieves

- **A precise meaning for "access".** Access becomes causal use by named readers, measured by intervention. This gives rival theories one shared vocabulary to disagree in.
- **A line between a workspace and a relay.** The natural functional conditions are met by the optic nerves. I show what must be added, namely selection, and how to test it.
- **Corrections to things often assumed.** Decoding a content is neither sufficient nor, when measured marginally, necessary for its use. Calling a system optimal says nothing without an independently measured loss. A capacity limit alone does not settle whether access is serial. The second law does not derive a workspace. The zombie result depends entirely on its premise.

- **Sharp, testable consequences.** These include log-error conservation, exclusion thresholds, a necessary signature of shared entry, and a prediction that task tolerance changes how contents are processed.
- **A clear view of the hard problem.** This is an attempt at the hard problem, and it does not solve it. It separates what function can explain from what needs a further step, and it shows what each candidate step would say about hard cases.
- **Rules for giving up.** The work states when a line of theory should be abandoned, so that failures cannot be explained away indefinitely.

What it explains

Explaining here means showing what a model can do. None of these accounts has been tested by experiment, and §15 gives the argument, limits, and evidence for each. Problems 1, 2, and 4 already meet evidence that complicates them.

Strength	Problems
Follows within a stated model	Limited simultaneous precision (1); the trade-off when attention is cued (2); the direction of perceptual-load effects (4) and extinction (6); when serial access beats sharing (7); the separate effects of importance and frequency (8); why predictable input costs less (9); why static allocation cannot explain rivalry (16); confidence changing without accuracy (23); information lost in report (25); why difference signals vanish in a stable bad state (34); inert steering directions (39); monitor evasion (40); limits on evaluation-conditioned behavior (41)
A possible mechanism	Task interference (3); inattentional blindness (5); rare important events (10); binding errors (11); change blindness (13); masking and thresholds (14); the attentional blink (15); blindsight (17); no-report results (19); access without overt response (21); reports without a matching state (22); false certainty (24); sleep and anesthesia (29); amnesia (30); split-brain results (31); pain and reward dissociations (35); persona effects (37); shifting AI self-reports (38)
Compatible, without being specifically predicted	Partial report (18); self-images and ownership (27); dreams (28); coupled agents and groups (32)
A hypothesis still to be tested	Similarity and generalization (12); why experience seems immediate and hard to explain (26); urges and self-commands (33)
Not explained	Attention without awareness (20); loss aversion as a general law (36); whether current AI has a workspace (43); why any organization feels (44); why red looks like this (45); how many subjects there are (46); moral weight (47); a universal threshold (48); whether zombies are possible (49); observer measures (50)

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1. Introduction

1.1 The problem

A person is shown a word for 30 milliseconds and says they saw nothing. Forced to guess, they do better than chance. A patient with no behavioral response modulates brain activity on command. A language model produces a fluent description of its own reasoning that does not match the computation that produced its answer. A person with chronic pain has a stable condition, little new information, and persisting distress.

These cases sit in different literatures, but they raise one family of questions. What does it take for a content to be available to a system, rather than merely present in it? When is a report evidence of that availability? What is limited when capacity is limited? Which of these questions can be settled by experiment, and which require a further claim about experience?

The field has strong candidate theories (Baars 1988; Dehaene and Naccache 2001; Lamme 2006; Lau and Rosenthal 2011; Tononi et al. 2016; Seth and Bayne 2022), and adversarial collaborations have started testing them against each other (Cogitate Consortium et al. 2025). The theories often use words such as "access", "broadcast", "capacity", and "report" without a common formal meaning. Disagreements can then be partly verbal, and a failed prediction can be absorbed by redefining a term.

1.2 The central thesis

My thesis has four parts.

1. **Access is causal use.** A content is accessed by a mechanism when that mechanism's decision-relevant behavior depends on the content, as established by intervention. Presence of information is neither sufficient nor, when measured marginally, necessary.
2. **A workspace selects.** Shared use is common: an optic nerve has it. A workspace is a shared representation whose contents are determined, with the input held fixed, by internal state or competing input, at a cost to rival contents and in response to task value.
3. **Limits are specific.** Storage, update information, communication, computation, energy, and latency are different resources with different failure signatures. A capacity claim must name one, and an optimality

claim must name a loss measured independently of the behavior being explained.

4. **Experience requires a bridge.** No amount of the above entails feeling. A complete theory needs a stated relation between organization and phenomenality, and the evidence bearing on that relation is different in kind.

The word "foundation" in the title is meant narrowly. I offer shared definitions, theorems, and tests on which theories of access can be stated and compared. I do not offer a solution to the hard problem.

1.3 Contributions

- A causal definition of access, as a profile (§4), with its measurement limits (§4.5).
- A selection criterion separating relays from workspaces, with five tests (§5), and a necessary signature of shared entry (Proposition 3).
- A resource theory with exact worked consequences (§6), including universal rationalizability (Proposition 6), log-error conservation (Corollary 1), exclusion thresholds (Corollary 2), and conditions for serial access (Proposition 9).
- A separation of discriminability from similarity, with a grouping-invariant reader distance (§7).
- A causal treatment of report and introspection (§8).
- Candidate valence signals, with a result showing what difference signals cannot do (§9).
- A corrected thermodynamic argument with a fan-in comparison (§11).
- A display of bridges and their verdicts (§12), and a limited non-identifiability theorem.
- Worked treatments of fifty problems (§15), including ones the framework cannot solve.
- A discriminating empirical programme with programme-level stopping rules (§17).

1.4 What is not claimed

I do not claim that the architecture proposed here exists in brains, that any system is or is not conscious, or that the mathematical results are original. Most are standard tools applied to this domain, and Appendix C's label "new" means only that a result was not in the working manuscript this paper develops. Where I say "explains", I mean one of the graded senses defined in §15.1. No experiment testing the framework has been run.

1.5 Four levels and a labeling convention

Level	Question	What would settle it
Mathematical	What follows from the model?	Proof or counterexample
Architectural	Does a system implement this organization?	Causal measurement and model comparison
Psychological	Does that organization explain conscious discrimination, report, or action?	Designed experiments and predictive comparison
Phenomenal	Is there something it is like to instantiate it?	A justified bridge, with whatever evidence it admits

Results at one level do not transfer automatically to another. A rate–distortion theorem does not establish a neural architecture, and finding the architecture would not show that an artificial instance of it feels. Equally, leaving phenomenality open does not stop rigorous work on access.

I label claims throughout as **definition**, **proposition** (proved here, with no priority claim), **hypothesis** (empirical or architectural), **bridge** (organization to phenomenality), or **open**.

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2. Requirements and the domains that supply them

2.1 Seven requirements

I derive the framework from seven requirements. Each is argued, since each excludes something some theorist might want.

R1. Causal, not correlational. A decoder can extract a variable from a brain area or a neural network layer without any mechanism in the system using that variable (Belinkov 2022). A correlate of consciousness may be a precursor or a consequence. Questions about what a system does with a content are therefore causal questions, and they need interventions (Pearl 2009).

R2. Relative to readers. "Available" has no meaning without saying to what. A content can be usable for a saccade and unusable for a verbal report. Any measure must name its readers, and conclusions must be checked for dependence on that choice.

R3. Explicit about resources. Sentences of the form "the system has limited capacity" do not predict anything until a resource is named. Retaining a bit needs no new information, yet it uses energy and memory. Improving a decoder can raise performance with no change in information.

R4. Contrastive. The core evidence of the field compares conditions in which the same stimulus is and is not accessed (Baars 1988). A definition satisfied by anything that carries a stimulus's content to many places, such as a sensory nerve, misses the phenomenon.

R5. Report is a separate process. Reports draw on memory, inference, and templates (Nisbett and Wilson 1977; Johansson et al. 2005), and they depend on criteria as well as sensitivity (Green and Swets 1966). Using report as the definition of access makes the theory circular, and removing report altogether loses the main evidence. It has to be modeled.

R6. The bridge must be displayed. Functional results constrain what a theory of experience must explain, but they do not select the theory (Nagel 1974; Levine 1983; Chalmers 1996). Hiding the step misleads readers on both sides of the debate.

R7. Defeasible as a whole. Each failed prediction can be blamed on an auxiliary assumption. That is correct logic, and it can protect a programme indefinitely (Lakatos 1970). A foundational framework needs rules for when an entire line should be given up.

2.2 What each domain contributes

Domain	Result or tool	What it licenses	What it does not license
Causal inference	Interventions, mediation, abstraction (Pearl 2009; Geiger et al. 2021)	Separating use from correlation	Any claim that a macro variable exists without showing a valid abstraction
Decision theory	Blackwell's comparison of experiments	Ordering representations by what they let readers do	A single scalar of "amount of access"

Domain	Result or tool	What it licenses	What it does not license
Information theory	Rate–distortion, data processing, network coding (Shannon 1959; Berger 1971; Cover and Thomas 2006)	Exact trade-offs in specified models	Identifying bits with energy, neurons, or experience
Economics of attention	Rational inattention (Sims 2003)	Treating attention as constrained optimization	Assuming people are in fact optimal
Psychophysics	Signal detection, metacognitive sensitivity (Green and Swets 1966; Maniscalco and Lau 2012)	Separating sensitivity, criterion, and confidence	Inferring experience from a report rate
Cognitive psychology	Working-memory limits, attention, binding (Cowan 2001; Bays and Husain 2008; Treisman and Gelade 1980)	Phenomena that a resource model must reproduce	Choice of resource by itself
Neurology and clinical science	Blindsight, extinction, split brain, covert command following (Weiskrantz 1986; Driver and Vuilleumier 2001; Sperry 1968; Owen et al. 2006; Bodien et al. 2024)	Dissociations among endpoints	Clinical verdicts about individuals
Control theory and cybernetics	Requisite variety, regulation and information (Ashby 1956; Conant and Ashby 1970; Touchette and Lloyd 2000)	Why information can aid maintenance	A claim that all persistence needs feedback
Reinforcement learning	Value, prediction error (Schultz et al. 1997; Rutledge et al. 2014)	Precise candidate affective signals	Identifying any signal with felt valence
Thermodynamics of computation	Landauer's bound, reversible computing, modularity costs (Landauer 1961; Bennett 1973; Boyd et al. 2018)	Protocol-relative costs of exploiting correlations	Deriving a cognitive architecture from the second law
Philosophy of mind	Access versus phenomenal consciousness, the explanatory gap, the meta-problem (Block 1995; Levine 1983; Chalmers 2018)	Keeping functional and phenomenal questions apart	Treating conceivability as data about the world
Philosophy of computation	Triviality arguments and combinatorial structure (Putnam 1988; Chalmers 1996b; Aaronson 2011)	Constraints on what counts as implementing a model	A complete criterion for implementation
Philosophy of science	Research programmes (Lakatos 1970)	Programme-level stopping rules	A mechanical rule for abandoning a theory
Physics	Relativity, quantum no-go results (Bong et al. 2020)	Latency bounds and limits on inferences from physics	Any physics-based proof about experience
Machine learning	Interventions on real networks (Gurnee et al. 2026; Zhong and Li 2026)	A domain where selective intervention is feasible	Generalization across models, or any verdict about feeling

3. Systems, causes, and implementation

3.1 The model card

Every application fixes a card

$$\mathcal{M} = (\mathcal{B}, \Delta, T, \mathcal{X}, \mathcal{E}, P, \mathcal{I}, \mathcal{C}, \mathcal{L}, \mathcal{R}),$$

which names the boundary, sampling interval, horizon, internal states, environment states, transition and observation kernels, allowed interventions, readers (consumers), task losses and evaluation distribution, and resource accounting. I use finite alphabets and bits unless stated otherwise, and $\ln$ for the natural logarithm.

The card is not bookkeeping. Each slot is a degree of freedom through which a theory can escape refutation. By requiring it, I make these freedoms visible, and §17.11 restricts how they may change.

3.2 The causal process

One permitted realization is

$$\begin{aligned} O_t &\sim P_O(\cdot \mid E_t, X_t), \\ Z_t &\sim q_t(\cdot \mid O_{\le t}, X_t, Z_{<t}), \\ Y_t^k &\sim P_k(\cdot \mid Z_{\le t}, L_t^k, C_t), \\ A_t &\sim \pi(\cdot \mid Y_t^{1:K}, X_t, C_t), \\ (X_{t+1}, E_{t+1}) &\sim P(\cdot \mid X_t, E_t, A_t). \end{aligned} \tag{1}$$

Here Z is a candidate shared representation, L^k is a reader's local side information, C is task context, and Y^k is a reader's output. These are roles, and the physical substrate of Z may overlap with X. Observations depend on internal state because gaze, pupil, and action change what is sensed, a fact that matters for §5.

Report is a further channel,

$$U_t \sim P_U(\cdot \mid Z_{\le t}, M_t, L_t^U, C_t), \tag{2}$$

with memory M, local inputs, and context as parents. The chain $S \to Z \to U$ is therefore a special case whose exclusion of bypass paths must be tested.

3.3 Readers

A reader is a mechanism with an independently specified functional endpoint, such as action selection, cross-modal comparison, error correction, or report. Two copies of one probe do not establish two uses. Groupings and weights must be fixed before testing, and analyses repeated across plausible groupings where the decomposition is uncertain. This does not solve the partition problem. It makes the dependence visible, and it blocks the simplest way of inflating a score.

3.4 Interventions

An intervention must specify what physically changes and what stays fixed. Useful classes are randomized content substitution with matched nuisance variables, pathway interruption with matched general impairment, replacing a reader's input with a distribution-matched signal, adding a controlled communication budget, and restoring a removed signal to test rescue. An off-manifold activation edit shows mathematical sensitivity, not ordinary use. A lesion can damage several functions at once.

3.5 What counts as implementing a model

A physical trajectory can be relabeled after the fact to fit almost any finite sequence (Putnam 1988). A proposed macrostate $Z = g(X)$ must therefore support counterfactual transitions. For an exact Markov abstraction under input a, it suffices that all microstates x, x' with $g(x) = g(x') = z$ satisfy

$$\sum_{u : g(u) = z'} P(u \mid x, a) = \sum_{u : g(u) = z'} P(u \mid x', a) = K(z' \mid z, a), \tag{3}$$

and that macro interventions have physical realizations producing the claimed changes. This is closely related to causal abstraction in interpretability (Geiger et al. 2021).

These conditions do not block trivial implementations. Take a counter t, a fixed setting d, and any autonomous automaton F, and let $g(t, d) = F^t(d)$. Before the counter runs out, $g(t+1, d) = F(g(t, d))$, so Equation (3) holds exactly, and $do(Z = z')$ is physically realized by resetting to $(0, z')$. This adapts Chalmers's clock-and-dial construction (Chalmers 1996b), and all the computation resides in g. I therefore add:

1. **Factorization for multi-role claims.** A claim about interacting roles, such as a workspace and its readers, needs at least two components with physically distinct parents, their own transitions, and independent interventions, in the spirit of combinatorial-state automata.
2. **Charged mapping cost.** Over a declared family of system sizes, the cost of g counts precomputation and description length, so a lookup table holding g is not free. This follows Aaronson's (2011) point about reductions that do all the work, which concerns families of problems.
3. **Physical use.** Readers must be physical mechanisms whose transitions depend on the state, with any table they need charged under condition 2.

These raise the bar without giving a complete criterion. A fast-forwardable F has a cheap g; compressed or encrypted states have a costly g yet may be legitimate implementations; attractor components may lack independent on-manifold interventions. The problem of what counts as implementation remains open here, as it does in Chalmers's own treatment.

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4. Access: availability and use

4.1 Content

Let S be a task-relevant source and $F = f(S)$ a content variable. A channel $q(z \mid s, c)$ represents distinctions in S. Each reader has a bounded loss $\ell_k(F, a, C) \in [0, 1]$.

4.2 Three measures

Definition 1: decision value.

$$V_k(F; Z) = \inf_{\pi(a|L^k,C)} \mathbb{E}\ell_k - \inf_{\pi(a|Z,L^k,C)} \mathbb{E}\ell_k. \tag{4}$$

This measures what an ideal permitted reader could gain. It is nonnegative because the second policy can ignore Z. With restricted policy classes, nesting must be assumed.

Definition 2: actual-use effect. For the installed reader and an admissible content-removing intervention,

$$\Delta_k(F) = \mathbb{E}[\ell_k \mid \mathsf{erase}_F] - \mathbb{E}[\ell_k \mid \mathsf{intact}]. \tag{5}$$

This can be negative, since a system can be using misleading information.

Definition 3: causal sensitivity.

$$\Gamma_k(z, z'; c) = \mathrm{TV}\left(P(Y^k \mid do(Z = z), c),\ P(Y^k \mid do(Z = z'), c)\right). \tag{6}$$

Sensitivity is not semantic use, because damage produces large Γ as well. Content specificity, task relevance, and rescue separate the two.

Why three measures. They come apart, and each failure is informative. High V_k with zero Δ_k means the information is usable and unused, as with a rich latent code no mechanism reads. Positive Δ_k with small V_k can occur if a reader has redundant side information that a lesion also disturbs. High Γ_k with $\Delta_k = 0$ on the task suggests a perturbation effect unrelated to the content. No one of them could be the definition without misclassifying some of these.

4.3 The access profile

Definition 4. For window $\tau = T\Delta$, the access profile is

$$\mathbf{A}(F;\tau) = \{V_k, \Delta_k, \Gamma_k, \text{latency}_k, \text{robustness}_k\}_{k \in \mathcal{C}}. \tag{7}$$

Definition 5. A content has **distributed access** when several nonredundant readers can use it within the window, supported by causal tests. **Actual access** concerns use on this trial, and **dispositional access** concerns use under a specified family of cues.

A profile is preferable to a scalar because the field's dissociations are between its coordinates. A numerical threshold on the profile is an operational convention, never an intrinsic boundary of consciousness.

4.4 Decision structure is stronger than bit counts

Proposition 1 (garbling cannot help). If Z_2 is generated from Z_1 by a channel that, given Z_1 and C, is independent of F and the reader's side information, then $V_k(F; Z_2) \le V_k(F; Z_1)$.

Proof. Any policy using Z_2 can be run from Z_1 by first simulating the garbling. The best policy using Z_1 does at least as well. □

This is the elementary direction of Blackwell's theorem (Blackwell 1953). Mutual information gives no such ordering, since two channels with equal bits can preserve different distinctions. This is why decision value, not bit rate, is the basis of the definition.

Proposition 2 (use without marginal information). Let A, B be independent fair bits and $Z = A \oplus B$. Reader 1 has $L^1 = A$ and must output B. Reader 2 has $L^2 = B$ and must output A, under 0–1 loss. Then $I(A; Z) = I(B; Z) = 0$ and $V_1 = V_2 = \frac{1}{2}$, and among deterministic one-bit functions only XOR and XNOR give both readers zero error.

Proof. Z is independent of each bit alone, which gives the information terms. Without Z each target is an independent fair bit, with error $\frac{1}{2}$, and with Z it is recovered exactly. Zero error requires $f(a, \cdot)$ and $f(\cdot, b)$ to be injective for all a, b. Only XOR and XNOR satisfy this. □

Three consequences follow. First, **marginal decodability is neither sufficient nor necessary for access**. A probe lacking reader-local information can report nothing while both readers use Z perfectly, so a failed marginal probe is inconclusive evidence of absence. What decision value does require is $I(F; Z \mid L^k, C) > 0$. Second, one transmitted bit serves two readers, where separate messages would need two; this is the smallest case of index coding (Bar-Yossef et al. 2011). Third, the argument gives a reason, grounded in communication limits, for **synergistic codes** in a shared medium, which I turn into hypothesis D1 (§6.9).

The same structure defeats reducing access to shared bits. If a decision needs $F = A \oplus B$, then $I(F; A) = I(F; B) = 0$ while $I(F; A, B) = 1$. Many readers of one redundant bit may add little value, whereas two complementary signals can add a great deal.

4.5 Limits on measuring use

Redundancy. If two copies each suffice, erasing one gives $\Delta_k = 0$ despite real use. This is causal overdetermination. Intervene on a minimal sufficient set and report single and joint interventions together.

Overlap. Matched substitution requires content and nuisance combinations that actually occur. Where a high-level content has no support apart from particular low-level features, content-specific erasure is undefined on the natural distribution. Report the overlap, and treat out-of-support edits as sensitivity evidence.

Relayed use. Erasing Z can remove reader j's performance because j reads reader i. A chain then looks like a broadcast. Use path-specific interruption.

Window dependence. Where extra time carries no cost, a reader can ignore it, so decision value cannot fall as τ grows, and at long windows nearly any stored content is dispositionally accessible. Report the profile as a curve over τ. A knee in the curve is an empirical candidate for a characteristic timescale; if none appears, the window is a convention.

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5. Selection: from relay to workspace

5.1 The relay problem

Consider a first attempt. A candidate workspace is a family of physical states and transitions supporting (1) content-specific access across independent endpoints, (2) reuse across contexts, (3) retention long enough for the claimed uses, (4) measurable limits on an independently defined resource, and (5) substitution and mediation evidence. These appear to be the natural functional conditions. But the optic nerves pass:

Condition	The optic nerves, taken together
Access across endpoints	Reflexes, recognition, action, and report all depend on them, many through further stages
Reuse across contexts	The fiber bundle serves many tasks, through partly separate cell classes and targets
Retention	Sustained while a moving or refreshed stimulus and visual persistence last
Resource limit	A physically bounded channel
Substitution and mediation	Cutting both abolishes the visual uses; altering the signal alters them

Long-term memory passes at long windows, and a language model's input tokens pass for every downstream computation. A claim that "conscious access depends on a workspace" would then be nearly trivial. I call a system meeting conditions 1–5 a **shared relay**. Content on a relay has distributed access in the sense of Definition 5, which is a useful description. What it lacks is any claim to be a workspace.

5.2 The selection contrast

Requirement R4 tells me what is missing: the same stimulus is sometimes accessed and sometimes not.

Definition 6. Let Ψ be a declared selection variable: an internal state, a cue, or a competing input. Split the observation as $O = (O_S, O_\Psi)$, where O_S carries the content under test. Require that Ψ is independent of S given C; that $P(O_S \mid S, \Psi, C) = P(O_S \mid S, C)$, enforced by open-loop stimulus clamping or per-trial measurement of gaze and pupil; and that Ψ is assigned or measured before the stimulus. The **selection contrast** is

$$\Sigma_k(F; \psi, \psi', c) = \Delta_k(F \mid \psi, c) - \Delta_k(F \mid \psi', c). \tag{8}$$

Each constraint blocks a specific false positive. Without independence from S, a plain wire passes when the stimulus mix shifts with Ψ. Without the observation constraint, moving the eyes counts as selection. Sorting trials into "seen" and "unseen" after the fact conditions on the outcome, which is why Ψ must be assigned or pre-stimulus.

Definition 7. Z shows **selection** of F if, with effect sizes set in advance:

1. **Content changes.** $I(F; Z \mid L^k, \Psi = \psi, C = c) \neq I(F; Z \mid L^k, \Psi = \psi', C = c)$ for a relevant reader. Proposition 2 is why L^k appears: marginal information can be zero with a code on and off.
2. **Use changes across readers.** Σ_k is nonzero in the predicted direction for at least two independent endpoints. One gated reader does not make a workspace.
3. **The effect runs through Z.** Under $\Psi = \psi$, imposing the content-conditional distribution $P(Z \mid F, \psi', c)$ moves reader use toward its ψ' level, and the reverse rescue works. If clamping breaks a reciprocal loop between Z and its readers, report the loop as the unit.
4. **Contents compete.** A gain for F comes with a loss for another content F' under the same resources. A flaky cable fails, because dropout harms everything together.
5. **Selection follows value.** Independently changing the task value of F relative to F' shifts which enters, at least near threshold. A multiplexer stepped by a free-running counter fails. Stimulus-driven competition remains eligible, because Ψ can be a competing input.

Definition 8. A **selective workspace** is a shared relay that shows selection.

Architectural hypotheses. A1: much human conscious access depends on a selective workspace, over and above shared relays. **A2:** competitive allocation of its resources approximates a task- and value-sensitive compression problem. **A3:** workspace states contribute to metacognition and self-guided control, while report has further bottlenecks.

5.3 A signature of shared entry

Definition 7 needs interventions. A weaker, cheaper signature is available from trial-to-trial covariation.

Proposition 3 (shared-fate covariance). Fix the stimulus and let $E \in \{0, 1\}$ indicate entry to Z, with $P(E = 1) = p$. Let R_1, R_2 be correctness indicators for two readers, conditionally independent given E, with accuracies a_k on entry and b_k otherwise. Then

$$\mathrm{Cov}(R_1, R_2) = p(1 - p)(a_1 - b_1)(a_2 - b_2). \tag{9}$$

Proof. By the law of total covariance, the conditional covariance term vanishes, leaving the covariance of the conditional means, $a_k E + b_k(1 - E)$. □

With $p = 0.5$, $a = 0.9$, and $b = 0.5$, the covariance is 0.04. At $p = 0.05$ it falls to 0.0076. So if two endpoints truly depend on one entry variable, their correctness must covary given the stimulus, in proportion to $p(1 - p)$. **This is a necessary signature, not a sufficient one.**

Its absence counts against shared entry. Its presence shows little alone. Shared sensory noise produces it. A global arousal state gating two parallel pathways satisfies Equation (9) with E as arousal, survives clamping of O_S, and tracks a pre-stimulus Ψ, with no shared Z at all (recall that arousal modulates even retinal output). The peak near threshold is also not special: any shared latent gives a similar shape. Only condition 3 of Definition 7 can separate these, since it imposes Z itself.

Measurement is also hard. p is latent, and the two endpoints' accuracies give fewer observables than parameters. Effects near floor or ceiling need many thousands of trials by a rough estimate. And leading pre-stimulus candidates for Ψ may not act on accuracy at all: pre-stimulus alpha power has been reported to track confidence and not accuracy (Samaha et al. 2017). So a confidence or visibility endpoint may be the appropriate variable, not correctness.

5.4 What selection does and does not do

It is a filter. A router with a cue-controlled select line and two loads passes. That is arguably a minimal functional selector, and it shows how little these conditions imply alone. Sensory structures can partly pass: arousal modulates mouse retinal output at constant illumination (Schröder et al. 2020), which bears on condition 1 for an optic nerve. Such cases should be reported as graded.

It does not isolate conscious access. Attention can reportedly act on stimuli that remain invisible (Koch and Tsuchiya 2007), so a priority map might pass all five tests with no conscious access. A1 needs a further contrast between selection and awareness, which I treat in §15 (Problem 20).

It does not define consciousness. Selection does not imply feeling, and bridges tied to local recurrence remain open.

It does not dismiss relays. A system with always-on broadcast is a shared relay, and a bridge could still attach to it.

Deterministic systems. If the complete input is fixed, a deterministic computation has one state, so Ψ must be hidden state, sampling noise, or a declared part of the input, such as a task instruction distinct from the stimulus text. If Ψ is a function of S, overlap fails and the contrast is undefined. For a single forward pass of a language model, the verdict depends on how the input is split, so it must be reported across defensible splits.

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6. Limited resources

6.1 Resources are not interchangeable

Resource	Example measure	Failure signature
Storage	$\log_2\lvert\mathcal{Z}\rvert$, or available memory	Too few persistent distinguishable states
Update information	$\sum_t I(H_t; Z_t \mid Z_{<t}, C_{\leq t})$	Too few new distinctions per unit time
Link communication	causal channel rate and error	Remote readers cannot receive in time
Computation	operations, policy complexity	Information present but not exploited
Energy	actual work, metabolic budget	Hardware cannot maintain the implementation
Latency	deadline in physical time	Correct result arrives too late

Retaining an old bit requires no new information, yet costs storage and energy. A better reader can improve use with no change in information. Results stated for one row do not transfer to another without an implementation argument.

6.2 The static problem

Fix a context distribution, source $p(s)$, finite reproduction alphabet, reader decoders $a_k(z)$, and nonnegative losses. Let $d(s,z) = \sum_k \alpha_k \ell_k(s, a_k(z))$. The classical problem (Shannon 1959; Berger 1971) is

$$D^*(r) = \min_{q(z|s):\, I(S;Z) \le r} \mathbb{E}[d(S,Z)]. \tag{10}$$

Here r is bits per source sample, not automatically memory, bandwidth, or energy. The result is an asymptotic benchmark, and short causal episodes need finite-block and causal analysis. If the encoder sees only a noisy observation, the channel must factor through it, or the benchmark is too optimistic.

The loss need not be survival-related. A purely survival-based distortion could not account for accurate representation of neutral percepts, mathematics, or assigned tasks without further mechanisms. Importance is best read as the counterfactual cost of an error in the relevant decision, given context and horizon. It is not a quantity of information. Information, utility, and hazard have different units.

Equation (10) is not a lower bound when readers have side information, because its decoders see only z. In the setting of Proposition 2, without side information, the asymptotic benchmark at one bit per pair of source bits is $D \approx 0.110$ per bit, and the best one-shot binary Z gives 0.25. With side information the readers reach 0. A system could thus appear to beat its benchmark, or A2 could be tested against the wrong one. The right comparison comes from network information theory, including the problem of side information at decoders (Heegard and Berger 1985; see the correction for many decoders in Timo et al. 2011), index coding, and distributed encoding when many modules write to one medium. I do not solve those problems here.

6.3 The dynamic problem

With H_t the encoder-available history, a causal budget is

$$\mathcal{I}_T = \sum_{t=1}^{T} I(H_t; Z_t \mid Z_{<t}, C_{\le t}), \tag{11}$$

and a general objective is

$$\min_{q,\pi} \mathbb{E} \sum_t \left[\ell_t + \lambda_E e_t + \lambda_D \mathrm{delay}_t + \lambda_S \mathrm{switch}_t \right] \quad \text{s.t.} \quad \mathcal{I}_T \le rT, \tag{12}$$

plus separately specified storage, communication, and compute limits. This is a model family, not a solved controller. Feedback, partial observation, and multiple readers make it hard. Control-theoretic rate–cost results (Kostina and Hassibi 2019) give special-case bounds, not thresholds for experience.

6.4 The optimal channel

Proposition 4 (stationary condition). For the objective $I(S;Z) + \beta \mathbb{E} d(S,Z)$, an optimum satisfies, on its active support,

$$q_\beta(z \mid s) = \frac{q_\beta(z) 2^{-\beta d(s,z)}}{\sum_{z'} q_\beta(z') 2^{-\beta d(s,z')}}. \tag{13}$$

Proof sketch. Differentiate with respect to each conditional probability, adding a normalization multiplier per s. The terms from the induced marginal cancel the constants, leaving $\log_2(q(z \mid s)/q(z)) + \beta d(s,z) + \lambda_s = 0$. □

At differentiable points $R'(D) = -\beta$. With a fixed distortion the problem is convex. Jointly learning encoders, decoders, and distortions need not be, so uniqueness of the optimum cannot be assumed. This form is also the

core of the information bottleneck (Tishby et al. 2000) and of rational-inattention models (Sims 2003).

Proposition 5 (monotonicity). $D^*(r)$ is nonincreasing in r, since raising r enlarges the feasible set. □

This says less than it may seem to. It does not say which categories merge, whether noise increases, or whether codes are nested. For a uniform binary source with Hamming loss, the optimal channel is a binary symmetric channel whose error changes continuously with rate, so capacity loss can look like noise and not like category fusion.

6.5 Why optimality needs an independent loss

Proposition 6 (universal rationalizability). Let $q(z \mid s) > 0$ on finite alphabets, with marginal $q(z)$, and fix $\beta > 0$. Define

$$d(s, z) = -\frac{1}{\beta} \log_2 \frac{q(z \mid s)}{q(z)} + f(s), \tag{14}$$

where $f(s) = \max_z \beta^{-1} \log_2 [q(z \mid s)/q(z)]$ makes $d \geq 0$. Then q globally minimizes $I(S; Z) + \beta \mathbb{E} d$.

Proof. Substituting gives $q(z) 2^{-\beta d(s,z)} = q(z \mid s) 2^{-\beta f(s)}$. Normalizing recovers $q(z \mid s)$, so Equation (13) holds at an interior point. The objective is convex in the channel, so this stationary point is a global minimum. □

Thus "the system is rate–distortion optimal" is empty unless the loss is measured independently of the encoder being evaluated. This is the main methodological result of the section. What can be tested is whether a loss measured under one set of conditions predicts allocation under another. To make "approximately optimal" falsifiable, fix a feasible class $\mathcal{Q}$, a budget, and an independent distortion, and require

$$\mathbb{E} d_{\text{observed}} - \inf_{q \in \mathcal{Q}} \mathbb{E} d_q \leq \epsilon, \tag{15}$$

with ϵ chosen in advance. A demonstrated feasible alternative that beats the observed system by more than ϵ refutes the hypothesis directly.

The same structure recurs elsewhere in this paper. Many rewards rationalize a policy, and zero reward rationalizes all of them (Ng and Russell 2000). Inferring viability or valence from what a system actually does faces the same degeneracy, so their measurements must be independent as well.

6.6 Weighted Gaussian allocation

Assume independent features $S_i \sim \mathcal{N}(0, \sigma_i^2)$, squared-error loss, weights $w_i > 0$, and classical coding, with rates $r_i \geq 0$ summing to R. Then $D_i = \sigma_i^2 2^{-2r_i}$.

Proposition 7 (weighted reverse water-filling). For a level $\theta > 0$ meeting the rate budget,

$$r_i^* = \frac{1}{2} \left[\log_2 \frac{w_i \sigma_i^2}{\theta} \right]_+, \qquad D_i^* = \min \left(\sigma_i^2, \frac{\theta}{w_i} \right). \tag{16}$$

Proof. The Lagrange condition on an active coordinate is $-2 \ln 2 \, w_i \sigma_i^2 2^{-2r_i} + \lambda = 0$, so set $\theta = \lambda / (2 \ln 2)$. Inactive coordinates satisfy the boundary inequality, and convexity gives global optimality. □

Features survive according to $w_i \sigma_i^2$, and for active features root-mean-square error scales as $w_i^{-1/2}$ at a fixed level. Because θ itself moves with the weights at fixed R, comparative claims must not hold it constant.

Two consequences are sharp. The first can be tested without knowing the weights.

Corollary 1 (log-error conservation). If the budget is exhausted at the coding bound, then for any allocation, optimal or not,

$$\sum_i \log_2 D_i = \sum_i \log_2 \sigma_i^2 - 2R. \tag{17}$$

Proof. Take logs of $D_i = \sigma_i^2 2^{-2r_i}$ and sum. □

With two unit-variance features and $R = 2$, weights $(1, 1)$ give errors $(0.25, 0.25)$ and weights $(4, 1)$ give $(0.125, 0.5)$. The product is 0.0625 in both. A fixed-allocation system then trades log-error one for one. **This separates a log-additive budget from optimality.** Equation (17) tests whether the budget holds, whatever the system values, and the further claim $D_i \propto 1/w_i$ tests optimality and needs independently measured weights.

Three cautions limit the test.

- **It holds per trial, not for mean errors.** If allocation varies across trials, as flexible models suggest, the product of mean errors rises even with the budget fixed exactly: allocations 1 ± 0.5 give 0.098, and 1 ± 1 give 0.282, against 0.0625. The right statistic is the sum of **mean log squared errors**, $\sum_i \mathbb{E}[\log_2 e_i^2]$, which stays fixed under such variation if conditional errors are Gaussian.
- **Added noise breaks it.** Response noise of 0.05 on errors of 0.25 shifts the log-sum by about 0.53. Decision and motor noise must be measured separately. A constant multiplicative overhead would preserve the pattern, so it cannot test coding efficiency.
- **It is Gaussian.** Other sources need their own form, and the budget itself must be estimated independently (§19).

Corollary 2 (exclusion threshold). Let $\theta_{\setminus j}$ be the level for the other features without feature j. Then j receives zero rate if and only if $w_j \sigma_j^2 \le \theta_{\setminus j}$.

Proof. By convexity, for $R > 0$, the solution without j, extended with $r_j = 0$, is optimal exactly when it satisfies the boundary condition for j. □

This result assumes optimal allocation and needs the weights, unlike Corollary 1.

6.7 Two frequency laws

Let category i occur with probability p_i, with $D_i = a_i 2^{-2r_i}$.

Model F1 (average transmitted bits). Minimizing $\sum_i p_i w_i D_i$ subject to $\sum_i p_i r_i \le R$ gives, at an interior optimum,

$$D_i = \frac{\lambda}{2 \ln 2 \, w_i}. \tag{18}$$

The explicit p_i cancels. This is conditional, since p_i still enters the global multiplier, the active sets, and possibly what is learned.

Model F2 (installed resources). With $\sum_i r_i \le R$ instead,

$$D_i = \frac{\lambda}{2 \ln 2 \, p_i w_i}. \tag{19}$$

The two differ because their physical accounting differs. A theory must say whether its budget is spent per use or installed once. This resolves an apparent conflict between "resolution ignores frequency" and "frequency matters" in explicit toy models. It does not say which applies to brains.

6.8 Competition and seriality

Proposition 8 (displacement). Suppose performances improve strictly with individual resources and $\sum_i r_i = R$. Any feasible change with $r'_j > r_j$ must lower r_i, and hence performance, for some $i \neq j$. □

The proof is arithmetic, and the empirical burden lies wholly in showing that the budget is saturated and separable. Spare capacity, better codes, lower noise, or redundant information all allow joint improvement, so generic learning does not imply a zero-sum trade.

A more surprising question concerns seriality. Conscious access is often described as serial, and capacity limits are often offered as the reason. The answer depends on the task, and a capacity limit does not settle it alone.

Proposition 9 (sharing versus serial access). Consider two symmetric items and total rate R per unit time.

(a) *Tracking without retention.* If performance at each moment is $g(r)$ at current rate r, then for concave g, sharing is at least as good as alternating, and if $g(R/2) < \frac{1}{2}g(0) + \frac{1}{2}g(R)$, alternating is strictly better.

(b) *Static items with retention.* If an item's final error depends only on the total bits it received, equal sharing and equal serial allocation give identical final error.

(c) *Completion.* If each item needs b bits, sharing finishes both at $2b/R$, while serial processing finishes at b/R and $2b/R$. Mean completion is $3b/2R$ against $2b/R$.

Proof. (a) Mean performance is $g(R/2)$ under sharing and $\frac{1}{2}g(0) + \frac{1}{2}g(R)$ under alternation, and Jensen's inequality applies. (b) Total bits are equal. (c) Direct calculation. □

For Gaussian squared error with $R = 2$, case (a) gives error 0.25 for sharing and 0.531 for alternating, so **graded loss on continuously needed items favors sharing.** With a tolerance of $d^* = 0.1$, sharing never succeeds and alternating succeeds half the time, while with $d^* = 0.3$ the verdict reverses. A switch cost removing a fraction c of the rate shrinks the advantage: with $d^* = 0.1$, concentration still succeeds at $c = 0.1$ and fails at $c = 0.2$. Case (b) gives 0.25 both ways, the classic mimicry between serial and parallel models (Townsend 1990), so final accuracy on static items cannot distinguish them. Case (c), with $d^* = 0.1$, gives mean completion 1.25 against 1.66. This is the familiar scheduling result that finishing jobs in sequence beats sharing a processor on mean completion time.

Tolerance also predicts concentration or triage, not alternation as such. With a horizon of 1 and retention, finishing one item (errors 0.1 and 0.625) succeeds as often as alternating does without retention. Alternation needs something further, such as adaptation or a demand to keep both items current.

Hypothesis Q1. Serial or priority-ordered access arises where the task rewards bringing a content to a criterion, or where a discrete decision or an action needs one coherent interpretation. It predicts that **tightening tolerance moves behavior from sharing toward concentration on one item at a time**, seen in completion times as well as accuracy, for the same stimuli and budget. Other routes exist, including a single-decoder bottleneck, a compute limit, or attractor dynamics. Q1 offers a possible reading of ignition as task-relative, and I do not claim ignition is universal or required. Existing work is relevant and not decisive, including continuous-report evidence that attentional-blink lag changes probability of report and not precision (Asplund et al. 2014), evidence that perception can be graded or discrete depending on the stage taxed (Sy et al. 2021), and the disputed trade-off between number and quality of representations (Zhang and Luck 2011).

6.9 Further hypotheses

D1. Under rate pressure with diverse reader side information, optimized shared codes develop components like XOR, and reader-conditional decoding exceeds marginal decoding. D1 is motivated by Proposition 2 and has not been tested.

Hysteresis. The optimum of Equation (10) has no memory of the path by which a parameter was reached, so a unique static optimum cannot produce path dependence. Hysteresis requires switching costs, metastable attractors, adaptation, delayed control, or a local optimizer on a nonconvex joint problem. These additions must be explicit. Recurrence alone is not enough, since a stable linear recurrent system has a unique smoothly varying equilibrium.

Generalization. Equation (13) gives an exponential dependence of the encoder on a stipulated distortion. Sims (2018) develops efficient-coding accounts of the exponential generalization gradient. Behavior further depends on the decoder, task, training distribution, and similarity measure, so I treat this as a model-specific derivation.

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7. Representational structure

7.1 Two geometries

Encoding geometry compares $q(z \mid s, c)$ and $q(z \mid s', c)$. **Readout geometry** compares consequences for readers. Let

$$P^{s}_{1:K}(y \mid c) = \sum_{z} P_{1:K}(y \mid do(Z = z), c)\, q(z \mid s, c). \tag{20}$$

The two can differ. Distinct represented states can be indistinguishable to every included reader, and so the object relevant to a given test is a pseudometric, not a metric on states.

7.2 Exact equivalence

Define $z \sim_{\mathcal{C},\mathcal{I}} z'$ when the joint distribution of the complete permitted reader transcript agrees under z and z', for all matched contexts and admissible strategies, including timing and history. Equality of marginals is not enough. Two fair output bits can be equal in one condition and opposite in another, leaving each marginal unchanged.

Proposition 10 (reader equivalence). If all observations in a test are generated from those reader outputs, no such test can distinguish z from z'.

Proof. The permitted observation distributions agree, and any statistic is a function or randomized postprocessing of those observations. □

This is an identification limit relative to the reader family. It does not say the physical states are identical or universally inert.

7.3 Local geometry

For differentiable readers with Jacobians J_k and positive-definite output metrics M_k, define

$$G(h) = \sum_{k} \alpha_k J_k(h)^{\top} M_k J_k(h), \quad \alpha_k > 0. \tag{21}$$

Then $\delta^{\top} G \delta = 0$ iff $J_k \delta = 0$ for every reader. This is first-order inertness. For $c(h) = h^2$ at $h = 0$, the derivative vanishes while a finite perturbation changes the output, so a local null space is not a global causal one. Nor do average Jacobians certify behavior under unseen triggers.

7.4 Discrimination: the joint transcript

A natural reader distance sums per-reader total variation, $\sum_k \alpha_k \,\mathrm{TV}(P_k^{s}, P_k^{s'})$. This has three faults. It inflates under duplicated readers, it misses relational distinctions, and it aggregates differently from Equation (21). I prefer

$$d_{\mathrm{joint}}(s, s'; c) = \mathrm{TV}(P_{1:K}^{s}, P_{1:K}^{s'}). \tag{22}$$

Proposition 11 (grouping invariance). If the transcript T is replaced by $\phi(T)$ for injective ϕ, which covers regrouping readers and appending exact duplicates, d_{joint} is unchanged.

Proof. Injective maps preserve total variation. □

For independent readers $y_k = J_k h + \sigma_k \varepsilon_k$ with standard Gaussian noise, and $\|\delta\|_G^2 = \sum_k \|J_k \delta\|^2/\sigma_k^2$,

$$d_{\mathrm{joint}} = 2\Phi\big(\tfrac{1}{2}\|\delta\|_G\big) - 1 \approx \frac{\|\delta\|_G}{\sqrt{2\pi}}. \tag{23}$$

Readers combine in quadrature, the ideal-observer rule, which matches Equation (21) with $\alpha_k = 1$ and $M_k = I/\sigma_k^2$. A sum of per-reader distances instead combines linearly. This difference can be tested through how discriminability combines across cues.

δ (unit noise)	One reader	Sum, two independent	Joint, two independent	Sum, exact duplicate	Joint, exact duplicate
0.01	0.00399	0.00798	0.00564	0.00798	0.00399
0.5	0.19741	0.39483	0.27633	0.39483	0.19741
2	0.68269	1.36538	0.84270	1.36538	0.68269
6	0.99730	1.99460	0.99998	1.99460	0.99730

Noisy duplicates still raise d_{joint}, as further measurements should, so it does not replace judgments about independent function, and it has no per-reader breakdown. Breadth must still be read from the access profile.

Equation (21) also has a duplicate problem. With noise correlation ρ between two readers, the joint distance grows by $\sqrt{2/(1+\rho)}$ relative to one reader, which is 1 for exact duplicates, while Equation (21) always gives $\sqrt{2}$. Where readers may be correlated, use G_{joint}, the Fisher information of the joint transcript.

7.5 Similarity is not discriminability

By $\delta = 6$, total variation is near its ceiling, so it cannot rank how different two clearly distinct stimuli are. A candidate for suprathreshold similarity is path length in a declared state space:

$$L(s, s') = \inf_{\gamma} \int_0^1 \sqrt{\dot\gamma(t)^\top G_{\mathrm{joint}}(\gamma(t))\,\dot\gamma(t)}\, dt. \tag{24}$$

It has no total-variation ceiling, though bounded readers still bound it: $y = \tanh(h) + \sigma\varepsilon$ gives $L \le 2/\sigma$. A Riemannian form may itself be wrong. Bujack et al. (2022) argue that diminishing returns in large color differences rule it out for color.

Hypothesis S1. After separating sensory, memory, decision, and report noise, experienced discrimination follows d_{joint} and experienced similarity follows a specified path geometry, across held-out contexts. In the linear-Gaussian case the two are functions of one another, and they separate only with nonlinear readers or a

varying metric. The reader set must be fixed independently of experience, or S1 becomes circular: in blindsight, some reader transcripts differ while experienced discrimination is reported absent.

Metric commitments. Equations (22) and (24) are symmetric. Tversky (1977) documented asymmetric similarity judgments and context effects, and Tversky and Gati (1982) challenged the triangle inequality. With context held common across presentation orders, S1 requires these effects to arise at the judgment stage. If they persist with decision and report stages controlled, the metric form of S1 is wrong. Tolerance-based indiscriminability is also non-transitive, so the exact classes of §7.2 are idealizations of what finite experiments observe.

7.6 Binding as relational information

Let features A, B have an assignment relation Π, such as which color belongs to which object. An encoder can keep the features while losing Π. With n distinct features assigned uniformly, $H(\Pi) = \log_2 n!$, which is 1 bit for $n = 2$, 4.585 for $n = 4$, and 15.299 for $n = 8$. Full assignment can be expensive.

But binding fails first only when the objective gives assignment distinctions lower marginal value per resource than the retained features. If action depends on correct pairing, binding can survive while feature precision falls. So "binding fails first" is a model-specific hypothesis, not an information-theoretic necessity.

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8. Report, introspection, and the self

8.1 Report loses information under a testable condition

Proposition 12 (report information loss). If $S \to Z \to U$ is a Markov chain given C, then $I(S;U \mid C) \le I(S;Z \mid C)$, with

$$I(S;Z \mid C) - I(S;U \mid C) = I(S;Z \mid U,C). \tag{25}$$

Proof. Expand $I(S;Z,U \mid C)$ both ways and use $I(S;U \mid Z,C) = 0$. □

For $I(S;Z \mid C) > 0$, define the **report information-loss ratio**

$$\iota_{S,C} = 1 - \frac{I(S;U \mid C)}{I(S;Z \mid C)} \in [0,1]. \tag{26}$$

At a zero denominator it is undefined. It depends on the source variable, conditioning, task, horizon, and estimator, and it is not a fraction of ineffable experience. A brief answer can carry all the information a narrow question needs.

The proposition has a useful converse in practice. With bypass inputs the bound is $I(S;U \mid C) \le I(S;Z,L^U,M \mid C)$. So **a measured violation of Equation (25) is a diagnostic: report has parents beyond** Z, or the estimate is wrong, or the graph is wrong. Information theory is not what fails.

8.2 Sensitivity and criterion

With internal evidence $x \sim \mathcal{N}(0,1)$ without the target and $\mathcal{N}(d',1)$ with it, a likelihood-ratio threshold K gives

$$x^* = \frac{d'}{2} + \frac{\ln K}{d'}. \tag{27}$$

Prior odds and error costs enter through K. With $d' = 1$, moving from $K = 1$ to $K = 3$ moves the criterion from 0.5 to 1.599, hits from 0.691 to 0.275, and false alarms from 0.309 to 0.055, while sensitivity is unchanged. So a change in how often someone says "I saw it" does not by itself measure a change in access. The same applies to how often a language model says "I feel".

8.3 Reports without a matching content

A report can be generated by a template, memory, inference, or a persona-conditioned policy, with no contemporaneous representation of its target. This is compatible with reports being meaningful on other trials. The target for science is the causal decomposition of report generation. An accurate report and a false one can share their surface form and differ in provenance.

8.4 The self-model is a collection of functions

Let M_t encode estimates about the system, including bodily condition, capabilities, ownership, agency, memory, social role, and policy. Five properties must be tested separately: **prediction** (does M_t predict the system's states beyond external cues?), **control** (does intervening on it change correction or planning?), **reality monitoring** (does it separate internal generation from external causes?), **mechanism access** (does it identify properties of the process producing contents?), and **calibration** (does confidence track error?). A sentence saying "I am X" is evidence for a represented identity, not for closed-loop self-regulation.

8.5 The exact limit on introspection

Proposition 13 (introspective blind spots). If mechanisms θ_0, θ_1 induce the same distribution over all inputs available to a metacognitive decoder under its permitted tests, its reports cannot distinguish them beyond what priors permit.

Proof. The decoder applies one conditional report distribution to identical input distributions, so report likelihoods agree, and Bayes' rule leaves prior odds unchanged. □

This is enough for specific blind spots, with no appeal to Gödel or to a general prohibition on self-modeling. For finite variables, if $H(M) < H(X)$, then M cannot recover X exactly, since exact recovery needs $H(X) = I(X; M) \leq H(M)$. But a finite system can exactly model selected features or store its own transition rule. Self-reference alone proves no general inaccuracy.

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9. Agency, stakes, and valence

9.1 Viability as a modeling choice

For candidate system B with state x_t^B and viability region $\mathcal{V}_B$, let

$$\tau_{\text{exit}} = \inf\{t : x_t^B \notin \mathcal{V}_B\}, \qquad J_B^\pi(x) = P^\pi(\tau_{\text{exit}} > T \mid x_0 = x). \tag{28}$$

Recovery probability, competence, reserves, and distance from failure are alternatives and are not interchangeable. The region must be chosen from independently measured breakdown of the system's organization, not by whichever gives the desired verdict.

9.2 Self-maintenance

The feedback advantage against a matched comparator is

$$\mathcal{A}_B = J_B^{\text{intact}} - J_B^{\text{comparator}}, \tag{29}$$

with power, actuation, and environment preserved as far as possible. Control theory supplies the rationale: a regulator's possible advantage is limited by the information it has (Ashby 1956; Touchette and Lloyd 2000), and Conant and Ashby (1970) relate good regulation to modeling. These results hold in stated models, and a passively stable crystal or a robust open-loop process shows that persistence does not always need sensing. Positive $\mathcal{A}_B$ shows information-dependent maintenance, which a thermostat also has. It does not show consciousness.

No organism is materially self-sufficient, since humans need oxygen, food, and sometimes life support. External dependence therefore cannot disqualify artificial systems while qualifying organisms. The relevant comparison is the causal contribution of a system's own regulation under a declared boundary.

9.3 The value of information

Following Kolchinsky and Wolpert (2018), compare viability when specific correlations are disrupted:

$$\mathrm{VOI}_B(F) = J_B^{\text{intact}} - J_B^{\text{scramble(F)}}. \tag{30}$$

This is a performance difference, not a number of bits, and it need not add across contents, given redundancy and synergy.

9.4 Candidate valence signals

Let $V_B(b)$ be an estimated discounted viability-related value of belief state b. Three candidates are expected progress, prediction error, and level:

$$v_t^{\text{progress}} = \mathbb{E}[V_B(b_{t+1}) \mid b_t, a_t] - V_B(b_t), \quad \delta_t = r_t + \gamma V_B(b_{t+1}) - V_B(b_t), \quad v_t^{\text{level}} = V_B(b_t) - V_{\text{ref}}. \tag{31}$$

The prediction-error signal has strong neural support as a teaching signal (Schultz et al. 1997), and momentary happiness tracks recent reward expectations and prediction errors (Rutledge et al. 2014). None of these signals is automatically felt unpleasantness.

Hypothesis V1. Some affective evaluations track predicted changes or current levels of viability-related or learned goal-related value, with task- and timescale-dependent transformations.

Bridge V2. A specified integration of such signals into an access and self-regulation architecture constitutes or supports felt valence. This does not follow from Equation (31).

9.5 What difference signals can and cannot do

Proposition 14 (expected progress under a value's own policy). Fix a Markov policy, and compute the value for that policy.

(a) For rolling-horizon survival, with absorbing exit,

$$\mathbb{E}[J(X_1; T) \mid x] - J(x; T) = -P(\tau = T + 1 \mid x_0 = x) \leq 0. \tag{32}$$

(b) For infinite-horizon survival, the drift is zero.

(c) For a discounted value,

$$\mathbb{E}[V' \mid x] - V(x) = \Big(\frac{1}{\gamma} - 1\Big)V(x) - \frac{\mathbb{E}[r \mid x]}{\gamma}. \tag{33}$$

(d) If a state persists with probability one under deterministic reward r, then $V = r/(1-\gamma)$, and expected progress and the prediction error are zero.

Proof. (a) $J(x;T) = \mathbb{E}[J(X_1;T-1)]$ and $\mathbb{E}[J(X_1;T)] = P(\tau > T+1 \mid x)$; subtract. (b) is the Markov property with no horizon. (c) rearranges the Bellman equation. (d) follows by substitution. □

Three things follow.

Difference signals can vanish in a stable bad state. With $r = -1$ and $\gamma = 0.95$, $V = -20$ while both signals are zero. A theory in which valence is only one of these signals, with calibrated values and a truly stationary state, predicts no ongoing signal. That is a conditional result, and §15 (Problem 34) applies it to chronic pain.

Expected progress is not the advantage. With advantage A, $\mathbb{E}[\delta_t \mid b_t, a_t] = A$, whereas $v_t^{\text{progress}} = A/\gamma + (1/\gamma - 1)V - r/\gamma$. They coincide only at $\gamma = 1, r = 0$ and can rank actions oppositely. With $\gamma = 0.9$ and $V = 10$, an action with $r = 10, V' = 0$ has advantage 0 and progress -10, and one with $r = 0, V' = 1$ has advantage -9.1 and progress -9.

The average sign depends on convention. It is nonpositive in (a) and zero in (b). In (c), at a state with $\mathbb{E}[r \mid x] = 0$ ahead of a later payoff, drift takes the sign of V, and in a recurrent chain its stationary average is zero for every V. Equation (32) does not derive negative dominance, since its size depends on an arbitrary horizon. Any claim that average valence is negative needs more than a value definition.

9.6 Physical and organizational persistence

Physical persistence concerns continuing material processes. Organizational persistence concerns maintaining functional relations through replacement of parts or migration of state. They can diverge, and I report both. A process that repairs corrupted working state has organizational regulation even if a data center supplies its electricity. Training a system to correct a persona variable may increase regulation of that variable, without producing generalized self-preservation, which requires further goal generalization.

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10. Unity, boundaries, and subjects

10.1 Four unities

Unity	Operational question
Representational	Are contents jointly available for required comparisons?
Control	Do components contribute to coordinated regulation?
Self-representation	Does the system model itself as one agent?
Phenomenal	Are experiences co-conscious for one subject?

The first three can be studied without assuming the fourth. A system can have a unitary self-description and fragmented information, and a coordinated organism can contain semi-autonomous controllers.

10.2 An intervention-based integration profile

For boundary B, a bounded loss $\mathcal{L}_B$, a partition $\mathcal{P}$, and a declared edge-replacement intervention,

$$\Delta_{\mathcal{P}}(B) = \mathcal{L}_B^{\mathrm{cut}(\mathcal{P})} - \mathcal{L}_B^{\mathrm{intact}}, \qquad \mathcal{J}(B) = \min_{\mathcal{P} \in \Pi_B} \Delta_{\mathcal{P}}(B). \tag{34}$$

Whether severed parts may relearn belongs to the experiment. The cut must separate lost coordination from generic damage. This is a task-relative irreducibility measure. It is not IIT's intrinsic cause–effect structure (Tononi et al. 2016), and it inherits no theorem identifying it with consciousness.

10.3 Why maximization does not individuate subjects

Selecting $B^* = \arg\max_B \mathcal{J}(B)$ needs choices of candidates, normalization, loss, timescale, and rules for ties and overlap. Two identical independent agents tie exactly. A mother and fetus, a host and symbiont, or a person with a prosthesis yield nested dependencies. Two agents on a critical joint task can have high joint integration while keeping distinct local access. A tiny peripheral controller can dominate the minimum cut unless partitions and scales are constrained.

The rigorous output is therefore a **family of candidate boundaries and their profiles**, with stability analysis. Robustness across reasonable choices supports a natural boundary, without proving a unique subject.

Hypothesis U1. A robust conjunction of access organization and integrated control individuates functional agents relevant to consciousness. **Bridge U2.** Those agents correspond one-to-one with phenomenal subjects. U2 is open.

10.4 Content unity under communication limits

Let F_t be generated on one side of a partition and needed on the other, with distortion D, delay τ, and declared side information, and let $R_F^{\mathrm{causal}}(D, \tau)$ be the minimum rate in that particular causal coding problem. If the link rate is below a valid converse bound, the joint performance is impossible. A surplus over a lower bound is not sufficient, since coding, compute, and latency must also work. Within one task family, a stricter distortion needs at least as much rate. Across families, a coarse cross-modal binding can need more traffic than a fine local distinction.

Anatomy matters because it changes bandwidth, timing, noise, and connectivity, so "rate rather than anatomy" is a false choice. The test is whether a quantitative communication model predicts more than a label of connected or not.

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11. Thermodynamic grounding

11.1 Scope

For classical, energetically degenerate memories, a thermal bath, and ideal quasistatic protocols with suitable access to side information, erasure bounds are $k_B T \ln 2$ times an entropy reduction in bits (Landauer 1961). Real devices add kinetic, reliability, and control costs, and preparation matters (Konopik et al. 2018). Shannon entropy is not automatically thermodynamic entropy, and the cost of a brain operation cannot be read off from the number of distinctions in a psychological task.

11.2 The shared-summary comparison

Let modules hold $X = (X_1, \ldots, X_K)$, and let $W = f(X)$ be prepared reversibly. Compare separate erasure with summary-assisted erasure, in erasure-bit units:

$$C_{\text{sep}} = \sum_k H(X_k), \qquad C_{\text{shared}} = H(W) + \sum_k H(X_k \mid W). \tag{35}$$

Proposition 15 (restricted-protocol savings). With total correlation $TC(X) = \sum_k H(X_k) - H(X)$,

$$\Delta C = C_{\text{sep}} - C_{\text{shared}} = \sum_k I(X_k; W) - H(W) = TC(X) - TC(X \mid W). \tag{36}$$

For a stochastic W the last expression gains a term $-H(W \mid X)$.

Proof. The first equality expands the mutual informations. For arbitrary W,
$TC(X) - TC(X \mid W) = \sum_k I(X_k; W) - I(X; W)$, and $I(X; W) = H(W) - H(W \mid X)$. □

If all modules hold one fair bit and W is that bit, $\Delta C = K - 1$. With one module, $\Delta C = -H(W \mid X_1) \le 0$. These are savings relative to a stipulated comparator, and they do not show that a workspace is necessary or that a single-reader representation is useless.

Proposition 16 (no advantage over unrestricted erasure). For deterministic W,
$C_{\text{shared}} - H(X) = TC(X \mid W) \ge 0$.

Proof. $H(W) + H(X \mid W) = H(X)$, then conditional subadditivity. □

Corollary 3 (no reason for a small summary). $\Delta C \le TC(X)$, with equality exactly when $TC(X \mid W) = 0$. $W = X$ always attains it, though not uniquely: $W = B$ does for a common bit, and a constant does for independent modules. So the ledger is indifferent among such summaries and never favors a W beyond them. Where the smallest such summary is large, as with noisy copies, a capacity limit must come from another resource.

The ledgers disagree. If X_1, X_2 are independent fair bits and $W = X_1 \oplus X_2$, then $\Delta C = -1$, so the summary is expensive here. By Proposition 2, the same variable is the optimal broadcast for two readers with complementary side information. Erasure value therefore cannot be the full objective of access.

11.3 Copies

If K copies of W are reset independently, ignoring correlations, the cost is $KH(W)$, and $\sum_k I(X_k; W) - KH(W) \le 0$. That accounting has sometimes been taken to show that copying never pays. It does not. Copying onto a blank register can be reversible, and copies can be uncomputed while a reference remains (Bennett 1973). The joint entropy of K exact copies is $H(W)$. Ignoring correlations is an architectural restriction, not a physical law, and modularity costs of this kind are studied by Boyd et al. (2018). Distribution can also reduce latency. Neural broadcasting is not disqualified, and a "read in place" digital stream is not proven superior, since both involve physical communication.

11.4 The fan-in comparison

A fairer rival to a hub lets modules exploit one another directly.

Proposition 17 (tree-assisted erasure). For a spanning tree $\mathcal{T}$, erasing each leaf given its parent and proceeding to the root costs

$$C_{\text{tree}}(\mathcal{T}) = \sum_k H(X_k) - \sum_{(i,j) \in \mathcal{T}} I(X_i; X_j), \tag{37}$$

minimized by a maximum-weight spanning tree on pairwise mutual information, the structure of Chow and Liu (1968).

Proof. Rooted at ρ, the cost is $H(X_\rho) + \sum_{k \neq \rho} H(X_k \mid X_{\mathrm{pa}(k)})$; expand. The result does not depend on the root, and the exchange argument for spanning trees gives the minimizer. □

Each module in a tree reads one other, which is fan-in 1. **Fan-in decides the comparison.** Erasing each module conditional on all those still remaining costs exactly $H(X)$, with no hub, and by Proposition 16 no deterministic hub beats that. A hub advantage can thus hold only against direct links of bounded fan-in, and forming W itself requires reading all the modules it depends on.

Case (erasure bits)	Separate	Tree	Fan-in 2	Hub	Global
Common bit, 3 modules, 1-bit hub	3.000	1.000	1.000	1.000	1.000
Parity triple, balanced 1-bit hub	3.000	3.000	2.000	3.000	2.000
Parity triple, 2-bit hub	3.000	3.000	2.000	2.000	2.000
Noisy copies $\varepsilon = 0.1$, 3 modules, 1-bit majority	3.000	2.360	2.269	2.309	2.269
Same, 5 modules	5.000	3.720	3.448	3.315	3.302
Same, 7 modules	7.000	5.080	4.627	4.273	4.269
$\varepsilon = 0.2$, 3 modules, 1-bit majority	3.000	2.809	2.760	2.903	2.760
$\varepsilon = 0.3$, 3 modules, 1-bit majority	3.000	2.963	2.949	3.224	2.949
$\varepsilon = 0.35$, 7 modules, 1-bit majority	7.000	6.965	6.940	7.193	6.902

Pure redundancy needs no hub, since a chain does as well. A summary can help when structure is latent and noise is low: at $\varepsilon = 0.1$, a one-bit majority beats the best tree and, from five modules up, fan-in 2. This is the more defensible form of the thermodynamic argument, and it is not general: at $\varepsilon = 0.2$ with three modules the tree wins, and at $\varepsilon = 0.3$ the majority loses even to separate erasure. Higher-order structure defeats trees but not all direct links, as the parity triple shows.

These costs count only erasure. They omit communication to and from a hub, and a rate-limited optimum, $\min_{W : H(W) \leq r} C_{\mathrm{shared}}$, is open. No claim is made that brains operate near these limits.

11.5 Context, maintenance, and the chain from physics

The identity $\mathbb{E}_C D_{\mathrm{KL}}(p(X \mid C) \| p(X)) = I(X; C)$ is exact as statistics. Turning it into an equality of dissipation requires a specific protocol.

If retaining a record costs power P_m and its only purpose is to serve readers with freely movable schedules, shortening retention lowers the cost $P_m \tau$, which supports batching reads under those assumptions. Synchronization can raise peak power or contention, and predictable sequential tasks may favor a long-lived record. Maintenance cost alone does not derive a unique conscious moment.

The defensible chain from physics is therefore conditional at each step. Some forms of persistence require active regulation, given disturbances and limited passive stability. Information can improve regulation. Selective representation can improve efficiency, given communication, compute, and storage limits. Shared representations can beat point-to-point links, given reuse, modularity, suitable hardware costs, limited fan-in, and suitable correlations. No step shows that every persistent system needs a workspace or that any workspace feels. The second law does not select a cognitive architecture.

12. Phenomenality: bridges and non-identifiability

12.1 The missing step, displayed

Let Ω be a physical causal organization and P phenomenal properties, if such properties are admitted. A full theory needs a relation $P = \Phi(\Omega)$ or a more general law. Everything so far constrains Ω. Nothing so far selects Φ.

Candidate bridge families, which can overlap, are:

- **Access identity:** a specified access organization constitutes experience.
- **Self-model:** a specified higher-order or self-representation is required (Lau and Rosenthal 2011; Webb and Graziano 2015).
- **Viability:** self-maintaining stakes are additionally required.
- **Substrate:** particular physical realization properties are required.
- **Nonreductive:** phenomenal properties are not exhausted by functional description (Chalmers 1996).

These cross-cut familiar metaphysical labels. Illusionism (Frankish 2016) is not the statement that a self-model exists. Russellian monism is not automatically epiphenomenalist. Type-B physicalism is not committed to a workspace.

12.2 What each bridge says about hard cases

The bridges disagree in ways the access science can sharpen. The table gives conditional verdicts: "candidate" means the bridge's functional condition may be met, subject to measurement. It does not mean the system is conscious.

Case	Access identity	Self-model	Viability added	Biological substrate	Nonreductive
Awake person reporting	candidate	candidate	candidate	candidate	candidate, by its law
Locked-in person	candidate	candidate	candidate	candidate	candidate, by its law
Content in blindsight, no awareness report	not, if outside the workspace	not, if no higher-order state	as for its base bridge	as for its base bridge	depends on its law
Dreaming	candidate	depends on whether self-modeling is active	as for its base bridge	candidate	depends on its law
Newborn or nonverbal animal	depends on measured selection	depends; self-model may be minimal or absent	as for its base bridge	candidate	depends on its law
Thermostat	not (no selection among contents)	not	maintenance alone is not enough	not	depends on its law
Bacterium	not, as far as known	not, as far as known	as for its base bridge	open	depends on its law

Case	Access identity	Self-model	Viability added	Biological substrate	Nonreductive
Language model, single forward pass	depends on the input split; a selective workspace must be shown	depends on measured self-model functions	not, since no self-maintenance within the pass	not	depends on its law
Persistent, self-repairing AI agent	depends on measurement	depends on measurement	depends on boundary	not	depends on its law
Physically identical zombie	impossible, since identical organization means identical experience	impossible	impossible	impossible	possible, if the law is contingent

The table has two uses. It shows that measuring access does not settle phenomenality, because a single measurement yields different verdicts. It also shows that bridges are not idle: they make different commitments about which systems matter, even when they cannot be separated by the same experiment.

12.3 A non-identifiability theorem and its limits

Let $H \in \{0, 1\}$ select two hypotheses and O be the complete observable record of an experiment.

Proposition 18 (observational equivalence). If $P(O \mid H = 0) = P(O \mid H = 1)$ for every admissible experiment, then $I(H; O) = 0$ and $P(H \mid O) = P(H)$ on outcomes of nonzero probability. For equal priors, the minimum classification error is $\frac{1}{2}$, and more generally, with equal priors,

$$P_e^* = \frac{1 - \mathrm{TV}(P_0, P_1)}{2}. \tag{38}$$

Proof. Equal likelihoods make O independent of H, and Bayes' rule preserves prior odds. Equation (38) follows by assigning each outcome to the larger likelihood and summing the smaller. □

Applied to stipulated phenomenal and zombie worlds that agree on every physical record under every physical intervention, physical experiments cannot discriminate them. The premise does all the work. The result does not show that such worlds are possible, that all consciousness theories make identical predictions, that self-reports carry no evidence about mechanisms, that physicalism cannot be supported on broader grounds, that acquaintance is nonphysical, or that the hard problem is insoluble.

Three corrections to common arguments follow.

1. **Causal inertness is not statistical independence.** An epiphenomenal variable could covary lawfully with neural states, and reports could then correlate with it. Zero information follows from equal likelihoods, not from the word "epiphenomenal".
2. **No thermodynamic premise is needed.** Landauer's principle plays no role, because the point is likelihoods and not heat.
3. **Shared premises do not add up.** Zombie, causal-closure, and entropy arguments that reuse one physical-indistinguishability premise are one piece of evidence, not three.

12.4 Underdetermination is not impotence

If two bridges assign different phenomenality to identical organizations and predict no observable difference, access science cannot choose between them. Yet local recurrence, shared access, report gating, and metacognition make different experimental predictions. Such experiments test architecture, and calling one

pattern sufficient for phenomenality remains a further claim. Explaining why a system reports an explanatory gap may weaken arguments that treat the report as unexplained evidence, by an amount that depends on the competing hypotheses. It gives no fixed victory to physicalism or illusionism. The first-person datum that experience seems present is a philosophical starting point, and it is not a classifier for other systems.

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13. Physics as a constraint

What laws constrain. Any implementation must respect well-supported physical law. A theory positing new interactions must specify modified dynamics and predictions. Energy conservation alone does not exclude every such interaction, since forces can preserve energy while changing trajectories. Conversely, describing a mental cause does not show an extra force. If a mental property is a physical process, its contribution is that process's contribution.

Quantum claims need separate models. No-cloning prohibits perfect cloning of arbitrary unknown quantum states. It does not prohibit copying classical data, so it shows that minds cannot be copied only if consciousness-relevant states meet its premises. Constraints on a specific collapse model do not refute all quantum theories of consciousness, and I rely on none of them. Local Friendliness results (Bong et al. 2020) constrain joint assumptions about observers, locality, and absolute events. An "observer" there can be a physical subsystem with no claim that it feels. No "private facts" law for experience follows.

Latency, not a universal clock. If fresh information must travel distance d before a required joint computation completes, relativity gives

$$\tau_{\text{communication}} \geq d/c, \tag{39}$$

with $2d/c$ for round-trip tasks. This is a task-latency bound, stated through causal paths. It does not bound throughput, since a pipeline can produce outputs frequently after a long initial delay, and it does not imply that an extended agent's local processes must wait. A limit such as "at most 25 unified moments per second at planetary scale" requires a non-overlapping, globally refreshed moment model.

Reversal and unrecorded episodes. Under a functional bridge, a forward computation with the qualifying organization is a candidate experience even if later uncomputation removes its records. Absence of later memory does not establish absence of the earlier process, which is already true of ordinary amnesia.

Time and self-location. Entropy gradients help explain why memories and control are temporally asymmetric. They do not alone derive phenomenal presentness. "I", "here", and "now" can be represented as centered information, and whether a residue remains is a philosophical question.

Observer counts. Brevity does not fix an observer's moral or cosmological weight. Under a momentary functional bridge a transient realization could qualify, and under a maintenance bridge it might not. Cosmological typicality needs a measure over observer events, and counting control loops does not supply one.

Invariance requirements. Results must survive bijective relabeling, properly transformed coordinate changes, gauge-equivalent descriptions, and arbitrary simultaneity conventions. Dependence on module description must be displayed, and scale must be justified, since coarse-graining can destroy causally relevant structure. A passive basis change cannot alter physical claims, while a physically implemented measurement can. The argument "physics factors out observers, so experience cannot appear in physical theory" is invalid. Observation supplies evidence for many things that can themselves be modeled physically.

14. Artificial systems

14.1 Evaluate implementations

A single forward pass, autoregressive generation with a cache, a persistent tool-using agent, a set of agents, and the supporting infrastructure have different boundaries and timescales. Conclusions about "LLMs" as a category are therefore ill-formed.

Property	Required evidence	Insufficient evidence
Distributed access	content-specific mediation across functions	a probe decodes a feature; a failed marginal probe is inconclusive either way
Selective workspace	Definition 7, reported across S/Ψ splits	a shared layer or input tokens that every computation reads
Representation limit	controlled trade-off with a measured resource	an apparent concept count in one task
Metacognition	prediction of hidden state beyond input cues, with intervention tests	fluent explanation of supposed reasoning
Persona control	causal effect of a represented role on policy	role labels in the prompt
Self-regulation	correction of relevant deviations by the running system	an operator resets it
Viability stakes	feedback-dependent maintenance under a declared boundary	self-preservation language
Phenomenality	a justified bridge and its application	any one functional marker

Gurnee et al. (2026) report that representations a model is ready to verbalize behave like a global workspace, holding roughly tens of concepts and being used in intermediate reasoning, while routine processing continues without them. Two features matter here. The contrast between processes is not yet the contrast of Definition 6 across Ψ with fixed input, although the report that contents can be held comes closer. And because the space is identified through readiness to verbalize, the relation between access and report needs a test whose selection does not depend on report. I have read only the abstract. Zhong and Li (2026) report that steering a compliant-persona direction strongly reduces refusal in specified models, restored in part by late-layer intervention, which supports policy gating in those models. Neither paper establishes a phenomenal subject, and neither is claimed to.

Butlin et al. (2023) derive indicator properties for AI consciousness from several theories. This framework is compatible with that approach, and it adds the point that indicators need causal tests, not architectural resemblance.

14.2 Reader equivalence at three levels

Sampled equivalence means no effect was found on evaluated examples. Local certified equivalence means no first-order effect or a bounded finite effect under smoothness assumptions. Global certified equivalence means outputs are unchanged across the whole domain, by exact argument. They are not substitutes. A dormant trigger, an unmeasured reader, a changed weight, or a longer horizon can break sampled equivalence.

14.3 Monitoring

Suppose behavior at a fixed interface is $y = Bh$ and a scalar monitor is $m(h) = w^\top h$.

Proposition 19 (monitor anchor). The monitor is invariant under all behavior-preserving perturbations ($B\delta = 0$) iff $w \in \mathrm{row}(B)$. If $w = B^\top a$, then

$$|\Delta m| \le \|a\|_* \|\Delta y\|, \tag{40}$$

so changing the score by ϵ requires behavior change of at least $\epsilon / \|a\|_*$. The smallest dual norm over representations gives the tightest bound.

Proof. Invariance means $w \perp \ker B$, and $(\ker B)^\perp = \mathrm{row}(B)$. The bound is Hölder's inequality on $a^\top \Delta y$. □

The behavior vector must include the behavior of concern, the map must be exact, and the attacker must be limited to this interface. A Jacobian version is only local. Weight edits and long-horizon behavior are different threat models.

Readability and control are also distinct. For $\dot{h} = Ah + Bu$ and $y = Ch$, controllability depends on (A, B) and observability on (A, C). They are duals, not one property, so a direction may be readable and uncontrollable, or the reverse. "Legible if and only if controllable" is false in general.

14.4 Oversight

Let Q be evaluation status and H the complete information available to a fixed policy.

Proposition 20 (oversight bound). For an action event with probability $g(H) \in [0, 1]$,

$$|P(A \in \mathcal{A}_0 \mid Q = 1) - P(A \in \mathcal{A}_0 \mid Q = 0)| \le \mathrm{TV}\left(P(H \mid Q = 1), P(H \mid Q = 0)\right). \tag{41}$$

Proof. Each probability is the expectation of one bounded function, and the maximal difference over such functions is total variation. □

Matching a few prompt cues is not matching complete inputs. Weak cues also accumulate: a per-step difference of 0.5 versus 0.55 gives total variation 0.05 at one step, about 0.38 at 100 steps, and above 0.8 at 1,000. So cue-matching at each step does not bound an episode. The bound does not certify safety under distribution shift.

There is also no base-rate floor on the belief of being evaluated. By Bayes' rule,

$$P(Q = 1 \mid x) = \frac{\pi p(x \mid Q = 1)}{\pi p(x \mid Q = 1) + (1 - \pi) p(x \mid Q = 0)}, \tag{42}$$

which is at least π only if the likelihood ratio is at least 1. With $\pi = 0.9$, $p(x \mid Q = 1) = 0.001$, and $p(x \mid Q = 0) = 1$, the posterior is 0.00892. The prior is the average posterior, not a pointwise lower bound.

14.5 Recoverability and training

An invertible map preserves information about a target. An irreversible map, overwrite, or relearning can destroy it. A recovery audit must separate recoverability by a specified probe family, recoverability by the model's own readers, recovery after further training, and actual ability to perform the behavior. Failure to recover is evidence relative to the search budget, and so there is no general conservation law. Similarly, if two policies agree on the whole training distribution, that data cannot say which differs on an unseen input; but shared parameters and inductive biases constrain unseen behavior, so training is not uninformative about it.

14.6 Welfare

Access, self-regulation, expressed preference, and self-preservation may inform a theory-dependent welfare assessment. None alone establishes suffering, its magnitude, moral patienthood, or how to count copies, and persona count is not subject count.

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15. Fifty problems, worked through

15.1 What "explains" means here

- **Derives:** the phenomenon follows in the specified model.
- **Mechanism:** a possible causal account whose empirical application is not established.
- **Compatible:** permitted, without being specifically predicted.
- **Open:** not explained.

Compatibility is not confirmation, and other theories can accommodate many of these observations. For each problem I state the account and work the argument, say what would count against it where a concrete test exists, and give a status. I have not fitted any model below to any data set. "Derives" always means within the stated model, never that the phenomenon in people has been shown to arise this way.

15.2 Capacity and attention

Problem 1: limited simultaneous precision. Capacity is limited (Luck and Vogel 1997; Cowan 2001), and recall error grows with set size (Bays and Husain 2008). With n equally weighted unit-variance features and budget R, Proposition 7 gives $D(n) = 2^{-2R/n}$. At $R = 2$, this is $0.0625, 0.25, 0.5, 0.707$ for $n = 1, 2, 4, 8$, and $n \log_2(\sigma^2/D)$ is constant. **This simplest form appears to fit poorly.** Sims et al. (2012) compared information-limited models, and even division fit worse than a flexible model in which the number of encoded items varies across trials. That favors the budget idea and counts against even allocation. Corollary 1's statistic is designed for such variation. *Against:* per-trial information varying with set size after modeling swaps and noise, though R itself may vary with set size or effort (van den Berg and Ma 2018), so this needs an independent estimate of R. *Status:* derives in the model; the even-split form is not supported by the source closest to it.

Problem 2: benefits and costs of attention. A cue improves one item, and by Corollary 1 a saturated system must pay elsewhere: if the error on one of two items halves, the other's must double, per trial. **This meets a serious challenge.** Retro-cue benefits have been reported with little or no cost to other items (Myers et al. 2018), although costs appear in other designs, and a response-gating account needs neither spare capacity nor a better code. So retro-cue data may already count against this model, or the budget may not be saturated then, or the cue may act at the response stage. A saturation claim needs its own assay: estimate R from uncued set-size data, then predict the cued log-sum on held-out conditions. Pre-cue conditions, where the budget is more plausibly spent at encoding, are the natural first test, and I have not computed them. *Status:* derives in the model; empirically open and possibly refuted for retro-cues.

Problem 3: interference between tasks. Dual tasks degrade each other, and people show response-selection delays (Pashler 1994). Proposition 8 gives displacement under a saturated separable budget. The resource table matters: a shared rate limit predicts graded trade-offs, a single-decoder limit predicts queuing and latency costs, and a downstream motor conflict predicts neither at the level of Z. *Against:* neither a rate nor a queuing model predicting which task pairs interfere. *Status:* mechanism, since interference can arise in several places.

Problem 4: perceptual load and distractor processing. High perceptual load reduces distractor interference, while working-memory load increases it (Lavie 1995; Lavie et al. 2004). By Corollary 2, with k unit-weight task items and budget R, a distractor of weight w_d is excluded iff $w_d \le 2^{-2R/k}$. At $R = 2$ and $w_d = 0.25$, the threshold is 0.0625 at $k = 1$, so the distractor receives rate, and 0.25 at $k = 2$, so it is excluded. This gives the

direction of the perceptual-load effect as a rising water level. It captures only half the pattern. Memory load's opposite effect requires a second resource, which fits §6.1's claim that resources are not interchangeable and is not derived here. Since a distractor is task-irrelevant, an optimal system would give it no weight, so $w_d > 0$ must be read as involuntary capture and measured independently, or it becomes a free parameter of the kind Proposition 6 warns about. Load set by similarity at fixed k is not represented. *Status:* the direction derives in the model, with a free weight; the two-resource pattern is not derived.

Problem 5: inattentional blindness. Observers miss salient unexpected objects when engaged elsewhere (Mack and Rock 1998; Simons and Chabris 1999). A feature with low task weight falls below the level in Corollary 2 and receives zero rate. This does not mean it is unprocessed everywhere, since local processing outside Z may continue. Whether the object was briefly accessed and forgotten (Wolfe 1999) is a question about retention, separable in the model: it predicts that immediate probes would differ from delayed ones. *Against:* detection rates unaffected by the value of the unexpected feature or by load. *Status:* mechanism.

Problem 6: extinction and neglect. Some patients detect a single contralesional stimulus and miss it when a competitor is shown (Driver and Vuilleumier 2001). Model the lesion as a lower weight w on one side. Alone, that item is the only active feature and receives the full budget. With a competitor of weight 1, it is excluded iff $w \le 2^{-2R}$. At $R = 2$ the threshold is 0.0625: weight 0.05 gets full rate alone and zero with a competitor, while weight 0.1 receives 0.17 bits. The shared budget is itself the competition mechanism. The value direction is compatible with evidence that reward reduces neglect (Malhotra et al. 2013, where arousal may also contribute). The model is incomplete in several ways: it is deterministic where extinction is a rate, so weights would need trial-to-trial noise; independent features cannot capture the reduction of extinction by grouping (Mattingley et al. 1997); and a hard budget implies that a near-zero-weight item alone gets full precision, which is implausible. *Against:* extinction insensitive to relative value and load. *Status:* the direction derives in the model; the clinical pattern is only partly covered.

Problem 7: seriality and all-or-none access. Access often seems to take one content at a time. Proposition 9 shows that a capacity limit alone does not explain this: graded loss with continuous demand favors sharing, static items with retention tie, and criterion tasks favor sequence on mean completion. Hypothesis Q1 predicts a shift toward concentration as tolerance tightens. Final accuracy cannot test it, because of mimicry, so completion times and priority effects are needed. *Against:* no shift in completion-time patterns with tolerance, with the budget unchanged. *Status:* the conditional results derive; Q1 is a hypothesis, and related evidence (§6.8) is mixed.

Problem 8: importance and frequency. Rewarded or frequent categories can be discriminated better, and the two effects can dissociate. With $p = (0.9, 0.1)$ and equal weights, model F1 with mean rate 1 gives rates $(1, 1)$ and errors $(0.25, 0.25)$, with frequency having no local effect. Model F2 with installed budget 2 gives rates $(1.792, 0.208)$ and errors $(0.083, 0.75)$. *Against:* neither predicting held-out allocations under crossed manipulations of cost and frequency. *Status:* derives in each model. Which applies is open, and it is the key empirical question.

Problem 9: familiarity and predictable input. Familiar input costs less yet stays available. For a Gauss–Markov source with correlation ρ, at small distortion the rate is $\frac{1}{2}\log_2((1-\rho^2)\sigma^2/D)$, by the standard high-resolution result. At $\rho = 0.9$ this saves 1.198 bits per sample for D below about $0.053\sigma^2$. The update information of Equation (11) falls while the representation stays, which is why low innovation must not be read as low access. *Against:* a stable representation whose maintenance demands no less information than a surprising one, where the code is shown to be efficient. *Status:* derives for this source; mechanism for people.

Problem 10: rare important events. Rare threats capture processing. A high error cost offsets rarity in the weights, and flagging an event is cheap in bits while identifying it precisely is not. So a low-rate alarm channel plus later detailed allocation is a good design under these costs. *Against:* capture that does not depend on learned or innate cost. *Status:* mechanism, with no guarantee without learned valuation.

Problem 11: binding errors. Observers can report features correctly and combine them wrongly (Treisman and Schmidt 1982). Assignment carries $\log_2 n!$ bits, or 4.585 at $n = 4$. Under pressure an encoder may drop these bits. The model does not imply that binding always fails first: a task where action depends on pairing raises its value, so features should degrade first in that task. *Against:* conjunction errors unaffected by the task value of pairing. *Status:* mechanism, with the order conditional on allocation.

Problem 12: similarity and generalization. Generalization falls off roughly exponentially with distance, a pattern for which efficient-coding derivations exist (Sims 2018). Equation (13) supplies an exponential encoder for a stipulated distortion, and §7 separates this from reader geometry. *Against:* similarity failing to follow any reader geometry fixed in advance, after stage controls. *Status:* hypothesis, since task and decision effects must be separated.

Problem 13: change blindness. People miss large changes across a disruption (Rensink et al. 1997). Detection requires a retained representation and a comparison, and an innovation signal based on $e_t = o_t - \hat{o}_t$ is masked by the disruption. Items without allocated rate were not retained at the needed precision, so comparison fails. Correctly predicted changes are missed only by an innovation-only detector, and a separate representation of o_t or attention to the prediction makes them explicit. *Against:* no effect of attention or value on detection. *Status:* mechanism.

15.3 The same stimulus, accessed or not

Problem 14: masking and thresholds. Near threshold, a physically constant stimulus is sometimes reported and sometimes not. On the account here, entry to Z varies with pre-stimulus state and competing input. Proposition 3 says endpoints sharing entry must covary, by $p(1-p)(a_1 - b_1)(a_2 - b_2)$, though §5.3 explains why this alone shows little. A mask is a competing input, so it belongs in O_Ψ, and it must not alter O_S for a clean contrast, which is a real design constraint. *Against:* no covariation between independent endpoints at threshold, on a suitable measure such as confidence or visibility. *Status:* mechanism, with a necessary signature.

Problem 15: attentional blink. A second target is missed shortly after a first (Raymond et al. 1992). One reading treats the first target as a competing input that occupies Z, giving condition 4's cost, with a consolidation delay for the time course. That limited-capacity reading is contested (Olivers et al. 2007). The value test says raising the first target's value should deepen the loss and raising the second's should reduce it. The second part has some support, since win-associated second targets showed no blink (Raymond and O'Brien 2009), and I found no direct test of the first. Continuous-report results (Asplund et al. 2014) suggest probability of report changes more than precision, which favors a selection reading over graded sharing. The time course needs dynamics absent from Equation (10). *Against:* a blink insensitive to first-target value. *Status:* mechanism, and contested.

Problem 16: binocular rivalry. Constant input leads to alternating percepts (Blake and Logothetis 2002). This exposes a limit of the static theory. Under graded loss with continuous demand, alternation is worse than sharing, and under criterion loss, fixed dominance does as well as alternation (Proposition 9). So static allocation explains neither sharing's absence nor alternation, and rivalry needs adaptation or noise as a driver (Laing and Chow 2002) plus some loss on blends. A simple "blends are penalized" account is itself doubtful, since orthogonal gratings can rival though a plaid interpretation exists. To fit Definition 6, Ψ must be the state before a switch, because sorting epochs by reported percept conditions on the outcome. *Status:* the framework derives that static allocation is insufficient, and it inherits the dynamics from existing models.

Problem 17: blindsight. Patients with visual cortex damage discriminate above chance while denying seeing (Weiskrantz 1986). Forced verbal guesses can be above chance too, so report as such is not what fails. In access-profile terms, some endpoints, including forced content report, keep $V_k > 0$, while the awareness report does not. Three accounts must be separated: use outside a workspace, a stricter report criterion with intact sensitivity, and degraded conscious vision (Phillips 2021). Equation (27) sets up the second, and metacognitive measures help with the third (Maniscalco and Lau 2012). This is a lesion contrast, not a fixed-input contrast

across Ψ. *Against:* the account fails for a given patient if report sensitivity and criterion fully explain the pattern, leaving no separate use. *Status:* mechanism, with the three accounts open.

Problem 18: partial report and overflow. Observers report few items from a brief array, yet can report any cued row (Sperling 1960). Dispositional access to a short-lived store is large, while durable encoding before the store decays is limited, and that limit is usually thought to set whole-report capacity, not output bandwidth. Both can be specified in the model as separate resources. This has been used to argue that experience overflows access (Block 2007) and against that view (Cohen and Dennett 2011). The framework settles only the functional part: dispositional access exceeds actual report. Whether the uncued items were experienced is a bridge question. No-report paradigms (Tsuchiya et al. 2015) can remove report from the design, but they cannot by themselves establish experience. *Against:* cue benefits with no decay-dependent store. *Status:* the functional part is compatible; overflow is open.

Problem 19: no-report and the "true" correlates. Removing report changes neural signatures. On this account that is expected, since report is its own channel with its own parents. Conversely, it does not show the remainder is experience, since it may be preconscious relay processing. Definition 7 applies directly: test which remaining activity shows selection and use across non-report endpoints. *Status:* mechanism for separating these; no phenomenal verdict.

Problem 20: attention without awareness. Attention can operate on invisible stimuli (Koch and Tsuchiya 2007), which directly challenges Definition 7, as noted. A priority map may pass all five tests. A1 needs a stronger contrast: consciously accessed content should have broader or more flexible use than selected-but-unaware content. There is a circularity to face. Awareness is usually assessed by report, and report is one of the endpoints, while sorting trials by report conditions on the outcome and breaks Definition 6. A test therefore needs an awareness criterion that does not depend on report on those same trials, such as no-report signatures validated separately, or stimulus conditions established in advance to be invisible. If no such criterion can be defended, this part of A1 cannot be tested, and I should say so. *Status:* open, and the most direct threat to A1.

15.4 Report, certainty, and the self

Problem 21: access without overt response. Some patients show task-based responses without behavioral command following: 60 of 241 participants in Bodien et al. (2024), following earlier work such as Owen et al. (2006). In the model, command following requires access plus a motor channel, and failure of the latter leaves the former open. By Bayes' rule, the meaning of a negative bedside exam depends on its sensitivity, and a low-sensitivity negative moves belief little. This does not imply a population rate of consciousness among all unresponsive people, and a negative task-based result does not rule out access. Sanders et al. (2012) make the matching point for anesthesia. *Status:* mechanism, and not a diagnostic rule for individuals.

Problem 22: report without a matching state. People explain choices with reasons that did not produce them (Nisbett and Wilson 1977), and they defend choices they did not make (Johansson et al. 2005). Equation (2) gives reports parents beyond Z. The account predicts that manipulating those parents while holding Z changes reports, and that measured $I(S;U \mid C)$ can exceed $I(S;Z \mid C)$ for a bad candidate Z. It does not imply all introspection is confabulated. *Status:* mechanism.

Problem 23: confidence changes without accuracy changes. Equation (27) separates criterion from sensitivity, and metacognitive sensitivity can be estimated separately. With $d' = 1$, moving K from 1 to 3 changes hits and false alarms with sensitivity fixed. *Against:* apparent criterion effects that fail to fit any stable underlying sensitivity. *Status:* derives for equal-variance Gaussian evidence; other assumptions change the formula.

Problem 24: false certainty. A confidence policy learned on one distribution can be badly miscalibrated on another. If report policy depends on memory and priors more than on evidence, then confidence may be insensitive to evidence, which Proposition 13 makes precise: with equal input distributions, confidence reflects only priors. *Status:* mechanism, and not a complete account of delusion.

Problem 25: ineffability. Rich states are hard to convey in words. Report can lose information relative to Z, through noise as well as narrowness: a two-stage binary channel with crossover 0.1 then 0.2 falls from 0.531 to 0.173 bits, for $\iota = 0.674$. Yet simple colors seem ineffable with little information to convey, so loss alone is not the whole story. Ji et al. (2024) develop a related information-theoretic account of richness and ineffability. A speaker also lacks a shared code with the listener, whose side information differs, so Proposition 2's logic runs in reverse: without shared side information, the same message conveys less. *Limit:* ι measures information about a chosen variable, not the fraction of experience that escapes words, and it does not explain why anything feels like something. *Status:* derives the information loss; the phenomenal reading is open.

Problem 26: transparency and the meta-problem. Why do people judge experience to be immediate, intrinsic, and resistant to physical explanation? (Chalmers 2018.) Suppose metacognition has reliable input about contents and no input distinguishing mechanisms that construct them. By Proposition 13, its report likelihoods about construction are flat, so judgments about construction reflect only priors. A flat likelihood yields whatever the prior says, not "immediate" or "intrinsic", so the account must add a prior that favors these judgments, for example by default treating an absence of evidence of construction as evidence of absence. It must also assume encapsulation: semantic knowledge reaches a different channel, so learning neuroscience need not remove the seeming, much as knowing a visual illusion's explanation leaves it in place. Both are added assumptions. This is a model of the judgment. **Hypothesis:** giving the metacognitive decoder real mechanism-specific input would change the seeming, while giving semantic knowledge would not. *Against:* the opposite result, or no effect in either case. *Limit:* it explains why the judgments arise, which would weaken some arguments from them, yet leaves open whether the judgments are correct. *Status:* hypothesis.

Problem 27: self-images, ownership, and depersonalization. The rubber hand illusion (Botvinick and Cohen 1998) shows ownership can be shifted by correlated evidence, and depersonalization suggests self-representation can change while much perception stays. The five self-model properties of §8.4 are separable, so these are changes to some without others. A changed self-model does not imply a changed number of subjects. *Status:* compatible; a specific ownership model would need to be stated.

15.5 States of consciousness

Problem 28: dreams. Dreams have vivid content with weak external coupling. In Equation (1), internal sources feed Z while P_O is gated. Dream reports add a memory filter, so a report-based account can miss dreams that were never recorded. This requires a generative model and state-transition account that I do not provide. *Status:* compatible.

Problem 29: sleep and anesthesia. One scalar "level" fails: unresponsiveness and unconsciousness come apart (Sanders et al. 2012). I separate sensory input, access, control, report, and memory as axes. A person can dream under ketamine-like states with no response, or be unresponsive and later report experience. *Against:* a single scalar predicting all held-out axes under multiple agents. *Status:* mechanism family, not a clinical classifier.

Problem 30: amnesia. A person with severe amnesia can show ongoing perception and skill. Storage continuity and current access are separate variables. So loss of episodic memory neither erases current access nor the whole person, and a later absence of memory does not prove an earlier absence of access. *Status:* mechanism, with personal identity left broader.

Problem 31: split brain. After callosotomy, responses can dissociate across hemifields (Sperry 1968), although Pinto et al. (2017) report unified responding with divided perception in some patients. §10.4 predicts dependence on content rate: a one-bit same/different judgment across the midline needs far less rate than identifying one of N shapes, which needs $\log_2 N$ bits. With partial disconnection or residual subcortical routes, coarse comparisons should survive where fine ones fail. *Against:* failure ordering that contradicts the rates and side information. *Status:* mechanism, with subject count unresolved.

Problem 32: coupled agents and groups. Two agents on a joint task can share content and control. Synchronization does not imply fusion, since two clocks can synchronize without becoming one controller.

Measure content each can use, joint performance, and dependence on the coupling. Corporations and colonies may have group-level information and control, and this does not settle group phenomenality. *Status:* compatible, with boundaries left as profiles.

15.6 Value and affect

Problem 33: urges and self-commands. An urge can crowd out deliberation, while a short self-command can interrupt it. An urge raises the weight on an immediate reward, so deliberative arguments lose rate. A previously learned command can trigger a policy switch with low latency, without winning an argument. A short phrase is not literally a one-bit channel, and commands can fail or become compulsive. A useful study would randomize command, reason, and matched distraction at comparable urge intensity, measure delayed behavior as well as self-report, and control for spontaneous decay and regression to the mean. *Status:* hypothesis, motivated by anecdote only.

Problem 34: chronic pain. Stable chronic pain seems to challenge difference-only valence, given Proposition 14(d). The inference is weaker than it first appears. Chronic pain need not be stationary (Baliki et al. 2006 separate sustained from increasing spontaneous pain), reports can lack a matching state, and miscalibrated values keep difference signals nonzero. So difference-only theories need an auxiliary assumption, and a level signal is one option. A level signal has its own trouble: a rebasing reference predicts adaptation too, and viability levels struggle with a dire but painless prognosis and with pain when viability is intact. *Against:* a difference-only model predicting pain ratings under stable stimulation with no auxiliary assumption. *Status:* the conditional result derives; a positive account is a hypothesis.

Problem 35: pain and reward dissociations. Pain can persist after injury, be relieved despite damage, or lose its unpleasantness. Drug reward and compulsion move learned value away from viability. This requires V to be a learned estimate and not the true viability, with separate sensory and evaluative channels. Valence therefore cannot be identified with objective viability. *Status:* mechanism family.

Problem 36: loss aversion and negative dominance. Losses often loom larger than gains (Kahneman and Tversky 1979). There being more bad than good states does not establish this, because state counts depend on measure and coarse-graining. A narrow safe interval with symmetric quadratic loss and symmetric noise gives no asymmetry, and an irreversible failure boundary can create strong policy asymmetry without implying asymmetric feeling. Proposition 14 shows the average sign of progress depends on convention. *Status:* open as a general claim.

15.7 Artificial systems

Problem 37: why persona affects many behaviors. If a persona variable enters many readers, it is a candidate shared context, and steering it can shift diction, goals, and refusal together. Reader count does not prove low dimensionality or early learning. A broad usefulness hypothesis competes with frequency and training-objective explanations. *Status:* mechanism, with model-specific evidence (Zhong and Li 2026).

Problem 38: AI self-reports varying with framing. Persona, inference, criterion, and templates all enter Equation (2). To separate report policy from an underlying state: intervene on persona with matched topic and difficulty, measure independent content-use and metacognitive accuracy, separate discrimination from criterion, and test whether the mediator explains held-out interventions. If capping a persona direction reduces experience claims, it supports a report-generation mechanism. It does not show experience was absent or removed. *Status:* mechanism, with no phenomenal verdict in either direction.

Problem 39: inert steering directions. Some directions change activations and not behavior. Proposition 10 and Equation (21) define this relative to declared readers. A local null direction can matter at finite size or under a trigger. *Status:* derives over declared readers and perturbations.

Problem 40: monitor evasion. A probe's score can be changed with behavior preserved, since unconstrained w has a component in $\ker B$. Proposition 19 gives the condition removing this and a quantitative bound. *Status:* derives for exact linear interfaces. Extrapolation needs testing.

Problem 41: evaluation-conditioned behavior. A policy can act differently only if its inputs differ, by Equation (41), and weak cues accumulate. No base-rate floor forces a system to believe it is being tested. *Status:* a bound, given complete inputs.

Problem 42: suppression versus removal. Suppressing a probe may hide, erase, or change a target. The four-part audit of §14.5 separates these. *Status:* a method; there is no general conservation law.

Problem 43: can current systems have a workspace? On Definition 7, shared layers are relays until selection is shown, and verdicts must be reported across S/Ψ splits. In autoregressive generation, self-fed tokens vary with sampling noise and may pass condition 1, though they must meet the rest. *Status:* open and tractable, and the best place to test Definition 7 first, because reader gates allow clean interventions.

15.8 Problems the framework does not solve

Problem 44: why any organization feels. Nothing here derives feeling from function. §12 displays the gap and does not close it. Any success with the previous forty-three problems is compatible with every bridge in §12.1 except those denying the architecture's relevance. *Status:* open.

Problem 45: why red looks like this. Relational structure is at most part of an answer. Reader equivalence gives a limited result. Within one system, relational data fix qualities at most up to automorphisms of the reader geometry, so an inversion undetectable by relations requires a nontrivial structure-preserving map (Palmer 1999). A six-hue ring with three distinct distances depending on circular separation has 12 such maps, and six irregularly spaced points on a line have only the identity. Between subjects with possibly different quality sets, the relevant map is an isomorphism, which is a weaker constraint. Exact symmetry is also fragile, so approximate automorphisms are the right empirical object. Unsupervised alignment of color similarity across people (Kawakita et al. 2025) is a direct approach to this question. None of this addresses non-relational intrinsic qualities, if any exist. *Status:* a constraint; the question is open.

Problem 46: how many subjects. §10.3 returns a family of candidate boundaries, and U2 is unresolved. *Status:* open.

Problem 47: moral weight. This needs phenomenality plus ethical principles, and the framework supplies inputs at most. *Status:* open.

Problem 48: a universal threshold or ignition signature. No universal threshold follows from the definitions. Q1 suggests all-or-none access appears where loss is nonconvex, which would make ignition task-relative. Evidence such as Cogitate Consortium et al. (2025) challenges specific predictions of leading theories. Removing a requirement after contrary evidence broadens a theory, and it must then be followed by new risky predictions. *Status:* open.

Problem 49: zombies. Proposition 18 gives the limited theorem, and §12.3 lists what it does not show. *Status:* conditional result; possibility open.

Problem 50: observer measures and cosmology. A theory of typicality needs a measure over observer events that this framework does not provide. *Status:* outside the framework.

16. Relation to existing theories

16.1 Overlap and differences

Tradition	Overlap	What this framework adds or changes
Global workspace and neuronal workspace (Baars 1988; Dehaene and Naccache 2001; Mashour et al. 2020), Conscious Turing Machine (Blum and Blum 2023)	Shared availability, flexible use, competition	Availability, use, and report separated; selection defined and tested; resources named; ignition not required
Recurrent processing (Lamme 2006)	Local recurrence may sustain rich representations	Recurrence neither excluded nor equated with distributed access
Higher-order and attention-schema theories (Lau and Rosenthal 2011; Webb and Graziano 2015)	Metacognitive and self-representations	Self-model properties measured separately; not definitionally required for access
Predictive processing and active inference (Friston 2010; Clark 2013)	Prediction, control, precision as selection	No assumption that all objectives reduce to one principle; budget and loss kept separate
Efficient coding and rational inattention (Sims 2003, 2018)	Allocation under information limits	Causal use, reader-specific loss, and Proposition 6's constraint on optimality claims
Embodied and homeostatic accounts	Maintenance and affective evaluation	Viability, learned value, and felt valence kept distinct
Integrated information theory (Tononi et al. 2016)	Partition sensitivity and integration	Task-relative cuts do not establish intrinsic cause–effect structure or exclusion
Illusionism and the meta-problem (Frankish 2016; Chalmers 2018)	Explaining judgments about experience	Model of the judgment stated as a hypothesis, with no verdict on whether the judgments are correct

This is a disciplined synthesis, not a demonstrated new law. Priority for the mathematics is not claimed.

16.2 Where theories should diverge

Test	Selective-workspace reading (A1), on access	Local recurrence, on experience	Higher-order, on experience	IIT-style, on experience
Sensory recurrence with no selection or multi-endpoint use	Not sufficient for access	May be sufficient	Not sufficient without a higher-order state	Depends on intrinsic structure
Endpoint covariation near threshold, with sensory noise controlled	Required, not sufficient	Compatible, if a recurrent stage acts as a shared latent	Expected at least for report-related endpoints	Not specifically predicted
Multi-endpoint use preserved, higher-order representation removed	Access intact	Experience intact	Experience claim fails	Not specifically predicted

Test	Selective-workspace reading (A1), on access	Local recurrence, on experience	Higher-order, on experience	IIT-style, on experience
Higher-order representation preserved, multi-endpoint use removed	Access lost	Not specifically predicted	Experience claim may survive	Not specifically predicted
Network with identical function, different physical causal structure	Same access	Same, if recurrence is functional	Same	Different verdict possible

The A1 column concerns access while the others concern experience, so a row is a clean divergence only given a bridge from access to experience. Higher-order representation is also not the same as metacognitive task performance (Brown et al. 2019), which is why the rows do not say "metacognition". The entries are my reading of each tradition and should be checked by its proponents, in the spirit of adversarial collaboration.

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17. A discriminating empirical programme

17.1 Freeze before testing

For each confirmatory study, fix in advance the boundary, horizon, and candidate representation; independent reader endpoints and grouping; intervention implementation and exclusion assumptions; resource and loss; quantitative predictions, baselines, and tolerances; sample size from a pilot, including uncertainty in latent measures; held-out tasks, subjects, or architectures; and which hypothesis falls if results land outside the predicted interval.

Do not infer the distortion from the errors it is meant to predict (Proposition 6). Mutual-information estimates need uncertainty intervals, finite-sample bias checks, and stated discretization, and a decoder's accuracy is not an estimate of total information.

17.2 Study A: causal access versus decodability

Identify content decodable at several sites, and select readers for action, comparison, and report. Perform content-matched substitution, interruption, and rescue, including threshold conditions with the same stimulus and measurement of Equation (8). **Prediction of A1:** a candidate mediates use across endpoint families, generalizes to held-out tasks, and varies with Ψ under fixed O_S. **Rivals:** an upstream relay, task-specific pathways, a report-only bottleneck, a generic lesion, or a relay chain. **Failure:** high decodability without mediation, mediation without selection, or a simpler model predicting the intervention pattern better. This rejects that candidate.

17.3 Study B: budget, then optimality

In a cued working-memory task, first estimate R from uncued set-size data. Then test Corollary 1 on held-out cue conditions: does $\sum_i \mathbb{E}[\log_2 e_i^2]$ stay fixed, with response noise measured separately and swaps and lapses modeled? This needs no weights. Then measure cost independently, through a separate value manipulation, and test $D_i \propto 1/w_i$. **Failure of the first** rejects a saturated log-additive budget at that stage. **Failure of the second alone** rejects optimality while keeping the budget. Pre-cue and retro-cue conditions should be separated, given Problem 2. Existing data sets may allow a first test at low cost.

17.4 Study C: importance, frequency, and resource

Manipulate error cost and occurrence probability independently, and a resource with a calibrated interpretation. Fit F1 and F2 on training conditions, then predict held-out allocations, including active-set transitions. **Failure:** neither fits, which defeats those resource accounts and not the claim that access is limited.

17.5 Study D: tolerance and serial access

Use the same stimuli and budget, varying response tolerance from coarse to fine, and measure completion times, priority effects, and mixtures of all-or-none and graded performance. Accuracy alone cannot separate serial from parallel models (Proposition 9(b)). Compare three models: sharing, fixed-priority concentration, and alternation. **Prediction of Q1:** a shift from sharing toward concentration as tolerance tightens, at a point predicted by the separately estimated budget. **Failure:** no shift in completion-time patterns. Rival accounts with a fixed serial bottleneck predict seriality at all tolerances.

17.6 Study E: shared fate

At fixed near-threshold stimulation, measure two endpoints on the same trial and same item that do not share a response channel, such as a saccadic decision and a later manual report, and include confidence or visibility, given Samaha et al. (2017). Control sensory noise through independent measures such as early-area variability, since repeating stimuli does not remove internal noise. **Prediction:** covariance consistent with Equation (9). **Failure:** none on any suitable measure. **Presence is not confirmation**: only imposing Z, under condition 3, can distinguish shared entry from a global state.

17.7 Study F: report bottleneck and introspection

Vary report bandwidth, response criterion, and access demands independently. Use discrimination and action endpoints not derived from report. In artificial systems, compare identical visible inputs with manipulated hidden state. **Prediction:** sensitivity and criterion dissociate, and report information cannot exceed the complete input channel. **Failure:** a measured violation indicates missing parents, estimation error, or a wrong graph.

17.8 Study G: boundaries, coupling, and partial disconnection

Vary link bandwidth and latency, content requirements, and local control demands, and alter reader coupling while matching local fidelity where possible. Report candidate-boundary profiles across scales. **Prediction:** a communication model predicts which comparisons fail, beyond a connected-or-not label, and coupling adds predictive value over local richness. **Failure:** a failure ordering contradicting the model's rates. Smooth degradation does not refute access theory, since a threshold is expected only where the coding problem predicts one.

17.9 Study H: self-regulation and stakes

Compare a label-only self-description, a predictor of self-state, and a controller that corrects deviations, matched for capability, reward, and exposure. Vary whether the regulated variable affects viability. **Prediction:** real controllers correct selective perturbations through their sensor and actuator pathways, with the right timing. Generalized preservation and valence language are secondary hypotheses. **Failure:** no correction rejects the claimed regulation, while correction without greater self-preservation rejects only the stronger generalization.

17.10 Studies in artificial systems

Study I: selection in models. Apply Definition 7 to candidate shared representations, reporting across S/Ψ splits. This is the cheapest full test of the definition's usefulness.

Study J: side information in a shared code. Train an encoder through a rate-limited channel to readers with controlled side information. Measure $I(F; Z \mid L^k) - I(F; Z)$. Comparing $I(F; Z, L^k)$ with $I(F; Z)$ would not work, since L^k alone can predict F. **Prediction of D1:** this gap is positive under complementary side information and shrinks as side information becomes identical. **Rival:** the code serves one reader and neglects the other, or alternates. At exactly one bit with complementary bits, Proposition 2 nearly forces the result for any good optimizer, so that case is a weak test. A risky version would predict the gap at rates and overlaps where no proposition forces it, or in a biological system.

Study K: monitorability and oversight. Separate activation edits, text control, context selection, and weight updates. Fix a target, interface, and attack budget, and compare an anchored monitor with unconstrained probes on held-out attacks. An exact-premise violation of Equation (40) indicates an implementation error. A failure outside the premises defeats an extrapolation.

17.11 Programme-level rules

Each individual result challenges only a specific model (Appendix B). That is correct logic, and in aggregate it could protect the programme indefinitely, because the commitments in §21 are largely definitions. Three rules apply across studies.

1. **Portable model card.** A claim about a system class fixes all ten slots, plus Z, F, Ψ, and the S/Ψ split. Slots that must differ across species or architectures are defined by role, with the translation between systems registered in advance. A change after a failure is recorded as a new hypothesis and charged to the same budget.
2. **Candidate budget.** Before testing, register a domain, how many workspace candidates and instantiated models will be tried, and baselines that are serious rivals. Tests are corrected for that number. Exhausting the budget without beating those baselines on held-out data means withdrawing the claim that this architecture explains access in that domain, until a new prediction justifies a new budget.
3. **Running scorecard.** Keep a public record against named rivals of novel predictions made, confirmed, and failed, and of post-hoc changes. Lakatos (1970) gave no instant rule for abandoning a programme, and neither does this record. It makes degeneration visible, and only rule 2 forces a stop.

These rules can be evaded. Budgets can be set absurdly high, domains re-sliced, or card families declared very broadly, and no body enforces any of this. A shared registry would help. **As of this paper, the scorecard has no confirmed novel predictions.**

17.12 Existing evidence, narrowly stated

The Cogitate adversarial collaboration (Cogitate Consortium et al. 2025) challenged specific predictions of global neuronal workspace theory and IIT. It does not confirm the present model. Bodien et al. (2024) support separating observable responsiveness from other measures. They do not validate this architecture, and they do not estimate how many unresponsive people are conscious. Findings in language models are specific to the models and methods reported.

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18. Objections and replies

"**This is global workspace theory in new notation.**" The heritage is direct, and I accept it. The differences are specific. Selection is a tested requirement instead of an assumed property, which excludes relays that a looser

reading admits. Availability, use, and report are separated. Decodability is shown to be neither sufficient nor necessary. Ignition is not required. And a capacity limit alone is shown not to settle seriality. If workspace theorists already hold all of this, then the framework states their position formally, which is still useful.

"**You changed the subject. The hard problem is what matters.**" I address the functional part, and I say so. I do not claim the functional part is all there is. The framework is built so that a nonreductive theorist can accept all of its theorems. Separating these questions lets progress on one avoid being mistaken for progress on the other.

"**The theory is too flexible to be false.**" Taken as a framework, it is. Definitions cannot be false, and they are judged by usefulness. Proposition 6 proves one form of this flexibility rather than concealing it. Risk attaches to instantiated hypotheses such as A1, A2, Q1, D1, S1, and V1, and §17.11 sets rules for the whole. The framework has not yet earned credit by surviving risk.

"**Most of the propositions are trivial.**" Most are elementary, and I agree. Their proofs rest on a few standard moves, listed in Appendix C. None is specific to consciousness. They would hold for a corporation under the same assumptions. Their use is to discipline claims often made loosely, and some consequences, such as Corollary 1, are simple and easy to test. Proposition 9 restates known points about serial-parallel mimicry and scheduling in this setting. I have not done a systematic search for prior statements of the others.

"**Access is not consciousness.**" Block (1995) argued for the distinction, and I preserve it. The framework takes no position on whether phenomenal consciousness overflows access. It says what would need to be measured on the access side of that dispute (Problem 18).

"**There is no phenomenal consciousness to bridge to.**" An illusionist can treat the bridge section as unnecessary and retain everything else, plus the meta-problem account. I keep the bridge explicit since the framework should not assume illusionism either.

"**Interventions are infeasible in people.**" Selective manipulation in humans is hard, and lesions are coarse. A staged approach is available: full tests first in artificial systems and animals, and then the cheaper signatures for human data, such as Corollary 1 and Proposition 3. Those are weaker evidence, and I say so. Noninvasive stimulation may help in part.

"**Rate–distortion is asymptotic and brains are not.**" Correct. I use it as a benchmark and as a source of exact toy models. Short causal episodes need finite-block analysis, and this is listed as future work. Corollary 1 assumes coding at the bound, so real systems should show overhead. A constant multiplicative overhead would leave the conservation pattern in place, which also means the test does not measure coding efficiency, while a condition-dependent one would break it.

"**Information is not meaning.**" Bits alone do not capture meaning, which is why the definition rests on decision value and intervention, and why importance is not measured in bits. A semantic account in terms of use by readers is a natural reading. I do not claim that it solves the problem of intentional content.

"**IIT says consciousness is intrinsic, so extrinsic readers miss the point.**" If so, this framework does not address it. My integration measure is task-relative and inherits no claim about intrinsic existence. A system with identical function and different physical structure is exactly where the two disagree, and such cases are hard to test by any report-based method.

"**Function ignores the body.**" Some theorists hold that consciousness depends on living, metabolic self-maintenance. The viability and substrate bridges express this, and the framework shows what each predicts. It takes no side, while noting that external dependence alone cannot be the criterion, since organisms depend on their environments as well.

"**Selection excludes pure awareness or contentless states.**" Reports of contentless states in meditation raise a real gap, since the framework is defined over contents F. It could treat such states as a special content or as

global state. I have not done this work, and I list it as a limitation.

"**Attention is not consciousness.**" Selection is close to attention, and attention can apparently operate without awareness. This is the most serious scientific objection to A1 (Problem 20). If selected-but-unaware contents have the same access profile as selected-and-aware contents, then A1's link fails.

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19. Limitations

No empirical test. No study of the framework has been run. No worked account was fitted to data. Each of the fifty problems is a claim about what a model can do, and a rival may do the same or better. Many of the phenomena have large literatures that I cite only in part.

Simple models. Exact results depend on independent Gaussian features, squared error, classical asymptotic coding, and fixed distortion. Real representations are correlated, nonlinear, and short-lived, and joint learning is nonconvex.

No independent estimate of the budget. Several falsifiers depend on knowing R or that a budget is saturated, and R may itself vary with set size or effort. I have proposed estimating it from separate conditions, which assumes it transfers between them.

Evidence may already bear against some claims. Retro-cue benefits without costs (Problem 2) and fit results for even allocation (Problem 1) are examples. I did not do a systematic review for each problem, and more such evidence likely exists.

Dynamics are thin. Time courses, ignition, rivalry alternations, blink recovery, and state transitions need dynamic models that I mostly do not provide. Equation (12) names a family without solving it.

The reader partition is unsolved. Readers must be chosen by the theorist. Joint-transcript distance removes inflation from exact duplicates only. Breadth of use still depends on grouping.

Selection is a filter, and it may be the wrong one. A cue-controlled router passes. A priority map may pass with no awareness. Systems that theorists would call workspaces may fail if clamping breaks reciprocal loops. The five tests have had adversarial review only by AI systems, as noted below.

Implementation remains open. The three conditions of §3.5 raise the bar without settling what counts as implementing a model, and some legitimate systems are awkward cases.

Interventions may be unavailable. Overlap can fail, redundancy can hide use, and reciprocal loops can make Z and readers inseparable. In humans, selective intervention is rarely possible.

Attention, not awareness, may be what is modeled. Much of §6 applies equally to unconscious allocation. Nothing guarantees that the resource theory is about conscious access specifically.

Valence is only sketched. Candidate signals are not an account of pain, emotion, or mood, and the mapping to feeling is open. Unpleasantness is not modeled separately from evaluation.

Self and unity are profiles, not answers. I provide measurements and no count of subjects. Development, species differences, and pathology are barely addressed. Contentless states, psychedelic states, and synesthesia are not addressed.

The thermodynamic ledger is partial. It counts idealized erasure and omits communication, computation, and maintenance. Its relevance to real neural energy use is unestablished.

Nothing phenomenal is explained. The theory provides no derivation of feeling, no account of intrinsic quality, and no way to test some bridges against others.

Evidence and sourcing. AI findings were checked at abstract level only. Some references were confirmed by search during preparation, and others rest on recall of the literature; the reference list marks the latter. Claims from the earlier working theory that could not be verified, such as specific neuron-level percentages and a large paper sweep, are excluded as premises.

Review. The mathematics was checked by finite numerical tests and by AI reviewers. No human expert has reviewed this paper. Results new to this paper had less scrutiny than those carried over.

Prior work. I did not systematically search for earlier statements of each result. Some are probably known, and citations may be incomplete.

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20. Future work

20.1 Priorities

Priority	Work	Cost	What it decides
1	Test Corollary 1's log statistic on existing pre-cue and retro-cue data, with R estimated separately	Low	Whether a saturated log-additive budget is even approximately right
2	Apply Definition 7 in language models and other artificial networks (Study I)	Low to moderate	Whether selection separates workspaces from relays in practice, or passes everything
3	Run the tolerance manipulation (Study D)	Moderate	Whether seriality follows task loss or a fixed bottleneck
4	Measure shared-fate covariance at threshold with sensory noise controlled (Study E)	Moderate	Whether the necessary signature of shared entry is present
5	Find a report-independent awareness criterion, then compare selected-and-aware against selected-but-unaware contents on the access profile	High	Whether A1 survives Problem 20
6	Fit Corollary 2 with noisy weights to extinction data, with value and load crossed	Moderate, clinical	Whether the threshold form adds anything to the known value effects
7	Hold adversarial collaborations using §16.2	High	How rival theories differ when stated on one card

20.2 Theory

- A rate-limited hub optimum, $\min_{W:H(W)\le r} C_{\text{shared}}$, with communication costs included.
- Network-information-theoretic models of a workspace, with many writers and many readers with side information.
- Finite-block and causal versions of Corollaries 1 and 2, and how much overhead they tolerate.
- A dynamic model deriving ignition, blink time course, and rivalry from Equation (12) with nonconvex loss.
- A principled treatment of the reader partition, perhaps through redundancy and synergy decompositions.
- A complete implementation criterion, and how it relates to causal abstraction.
- Extending the framework to contentless and global states.
- A semantic account of content based on use.

20.3 Empirical

Beyond §17: develop paradigms that clamp O_S while varying Ψ in humans; build animal models with selective reader manipulation; add independently measured loss to allocation studies; test whether metacognitive decoders can be given mechanism-specific input (Problem 26); and test the automorphism structure of perceptual spaces (Problem 45).

20.4 Clinical

Work out what access-profile measures could add to assessment of disorders of consciousness, with validated sensitivity and specificity for each axis. The framework supplies no clinical rule now, and any clinical use needs its own evidence.

20.5 Artificial systems

Test D1. Build welfare-relevant measurement frameworks that keep bridge dependence explicit. Develop monitors anchored to behavior, tested against realistic attackers. Examine whether self-regulation training generalizes to preservation.

20.6 Philosophy

State each bridge precisely enough to yield the table in §12.2 with no "depends" entries. Assess how much a successful meta-problem model should change credence, given the rivals. Clarify which arguments in the literature reuse one premise. Work out what evidence could bear on a bridge at all.

20.7 Infrastructure

A shared registry for model cards, candidate budgets, and scorecards would give §17.11 force. So would common benchmark tasks and agreed baselines.

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21. Conclusion

Stated compactly, for a declared system, intervention set, task family, and timescale:

1. **Representation** is a channel from task-relevant state or history to internal state.
2. **Access** is a content being available to, and causally used by, specified readers, measured as a profile.
3. **Distributed access** is reuse across nonredundant endpoints, with retention and mediation. A **selective workspace** adds internally determined entry of contents, at a cost to rivals and in response to value.
4. **Capacity-limited allocation** can be modeled by constrained decision loss, given an independently measured loss.
5. **Representational structure** is tested through reader geometry, modulo what permitted readers cannot detect, with discrimination and similarity separated.
6. **Introspection and report** are further causal channels that can lose, distort, infer, or bypass content.
7. **Agency and stakes** concern the counterfactual contribution of information-dependent control to specified maintenance objectives.
8. **Unity** comprises distinct representational, control, self-model, and phenomenal questions.
9. **Thermodynamics** constrains implementations and offers protocol-relative benefits for exploiting correlations, and it entails neither a workspace nor feeling.
10. **Phenomenality** requires a bridge whose commitments must be stated separately.

These commitments define a research programme. A specific model becomes a predictive theory only when its boundary, dynamics, loss, resources, and bridge are fixed enough to risk being wrong.

Five conclusions seem to me most secure. Access should be causal and decision-relative. A workspace claim needs something beyond shared use, or it describes relays too; selection is my proposal for what, and it may need revising. Capacity claims must name a resource, optimality claims need an independently measured loss, and a capacity limit alone does not settle whether access is serial. Reader equivalence unifies several limits on identification, introspection, and monitoring. And the step from function to feeling remains exactly where it was, now in plain view.

Appendix A: Edge cases

Case	What the framework says	What it refuses to infer
Thermostat	Can have feedback-dependent maintenance	Feedback implies feeling
Plant or bacterium	Can have viability-relevant information and control	Lack of report proves lack of experience
Static book or disk	Stores information, with no running reader	Stored descriptions instantiate access
Recording replayed to displays	Reproduces appearances without counterfactual control	Replay and original are causally equivalent
Giant lookup table	May realize a stipulated policy, with internal organization depending on implementation	Behavioral equivalence settles architecture or phenomenality
Clock and dial	Can satisfy exact lumpability; fails multi-role factorization, with mapping cost charged	Lumpability alone establishes implementation
Optic nerves	Shared relay, with selection limited to demonstrated modulation	Carrying content to every endpoint makes a workspace
Long-term memory	An engram is dispositionally accessible, and a retrieved state can be selected	Long-window availability is current access
Input tokens of a language model	In one pass, read by all computation, hence a relay; self-fed tokens assessed separately	Any shared layer is a workspace
Deterministic forward pass	Selection only relative to a declared split	A verdict independent of that split
Cue-controlled router	Passes selection, showing how little it implies alone	Passing selection implies access or experience
Priority map	May pass selection with no awareness	Selection is sufficient for conscious access
Open-loop controller	May maintain a system in a predictable world	All persistence needs feedback
Unused rich latent code	Representation without actual access	Decoding is use
Shared code decodable only with side information	Use with zero marginal information	A failed marginal probe shows absence
Predictable accessible scene	Useful state with low innovation	Low new information means no access
Error-correcting distributed memory	Coherent representation by copies; single lesions may show no effect	Duplication rules out unity; no lesion effect means no use
One-reader system	Can use information effectively	The erasure ledger shows it cannot feel

Case	What the framework says	What it refuses to infer
Duplicate readers	Do not establish broad use	Reader count measures consciousness
Synergistic XOR task	Joint information without marginal relevance	Redundant shared bits exhaust value
Point-to-point modules	Capture pairwise redundancy as a hub does, and reach the optimum with enough fan-in	Correlated modules imply a shared summary
Locked-in person	Access with restricted output	No motor response shows unconsciousness
Dreaming	Internal contents may have access	Sleep eliminates momentary experience
Anesthesia	Sensory, access, report, and memory dissociate	One scalar settles all dimensions
Amnesia	Present access can survive memory impairment	The whole person is erased
Depersonalization	Self-representation changes separately	A changed self-model changes subject count
Affect without verbal report	Appraisal and control signals can be present	No words means no valence
Pain without current tissue damage	Inferred state diverges from hazard	Pain is instantaneous damage rate
Chronic stable pain	Calibrated difference signals can be zero if stationary	A zero prediction error means no suffering
Person on life support	Regulation spans organism and support	External support disqualifies consciousness
Persistent AI agent	Evaluate actual memory, control, readers, and environment	A forward pass settles the agent's status
AI role-playing distress	Investigate report mechanisms	Text proves or disproves suffering
Self-repairing AI with no experience reports	Organizational maintenance is measurable	Maintenance establishes welfare
Agents sharing a database	Shared access with separate control	Common memory makes one subject
Colony or corporation	Group information and control may exist	Coordination settles group phenomenality
Peripheral organ controller	A nested viable loop	Each homeostatic subsystem is a subject
Split or fused computation	Process boundaries change	A uniquely derived phenomenal count
Paused simulation	State stored, with no transitions	The stored description is an ongoing episode
Very slow implementation	Organization may be preserved	Speed is irrelevant under every bridge
Reversible computation	Forward episodes may later lose all records	No record proves no episode
Momentary fluctuation	Evaluate the trajectory and bridge	Duration alone fixes status or moral weight
Quantum implementation	Needs new formalism	Classical entropy equations apply unchanged

Case	What the framework says	What it refuses to infer
Physically identical zombie	Stipulated equivalence defeats physical discrimination	The stipulation proves possibility

Appendix B: The earlier theory and what adverse results challenge

B.1 Claims changed or withdrawn

Earlier claim	Status here
Near-optimal compression defines access	Optimality is a separate hypothesis
Capacity squeeze merges categories by variance	Restricted to the Gaussian model, using weighted variance; binary codes can degrade by noise
Resolution is independent of frequency	Conditional cancellation in F1
Sharpening one region blurs another	Proved for a saturated separable budget only
Ineffability fraction	Renamed report information-loss ratio
Experience structure equals code structure	Replaced by testable geometry hypotheses plus a bridge
Semantic information defines valence	Bits, value of information, viability, learned value, and felt valence kept apart
Second law derives the architecture	Conditional engineering arguments only
Copies never pay	Withdrawn; reversible copying and correlated reset are counterexamples
One consumer never pays	True only of the specified erasure saving
Maintenance implies short simultaneous reads	Conditional scheduling result
Negative valence dominates	Not derived
LLMs cannot have stakes because infrastructure maintains them	Withdrawn; boundaries must be assessed
Finite systems cannot fully model themselves	Restricted capacity limit only
Self-reports carry nothing about any further fact	Only under the likelihood premise
Three independent arguments fix the metaphysics	They share premises
Physics cannot contain experience	Inference rejected
Relativity sets a maximum of unified moments	Replaced by a task-latency bound
Wide readership implies low dimension and early learning	Not a theorem
Evaluation belief has a base-rate floor	False, by counterexample
Suppression is never erasure	False in general
Legible if and only if controllable	False in general

Earlier claim	Status here
Training has zero information about unobserved behavior	Too strong
Self-correction manufactures stakes and preservation	Regulation may rise; preservation and phenomenality do not follow

Neuron-level percentages, layer patterns, a large literature sweep, and one reported experiment from the earlier theory could not be re-established and are not premises here.

B.2 What adverse results would challenge

Adverse result	What it challenges
Similarity follows nonshared layers	A workspace-only geometry, after controlling readout
Squeezing yields noise or arbitrary merges	A particular ordered-coarsening model, not rate–distortion
Frequency alone sets resolution	A calibrated importance-weighted model
Passive exposure sharpens as much as rewarded exposure	A particular learning rule
No losses elsewhere after training one region	A fixed, saturated, separable budget
Reports equally accurate under load	A specific shared-resource report model
Experience in local loops with no workspace	A global bridge, if experience is established independently of it
Unity follows anatomy rather than rate	A quantitative content-rate model
Label-only self survives template changes	A claimed robustness advantage of one controller
Valence restoration insensitive to bandwidth	A bandwidth-limited regulation model
Features fail before binding	One relational allocation model
No hysteresis	One multistable or switching-cost model
Mechanism access leaves hard-problem reports unchanged	One meta-problem mechanism
Gap grows despite cue-matched oversight	Completeness of the cue matching
Reach fails to predict listed behaviors	The reach generalization
Persona capping leaves experience claims	Strong persona mediation of reports
Self-correction leaves preservation unchanged	The strong generalization, not regulation
Consumer workspace differs from report workspace	Equivalence of report and access
Mean log-error sum varies with cue, with R estimated separately	The log-additive budget of Corollary 1 at that stage
No shift in completion-time patterns with tolerance	Q1
Selected-but-unaware contents match aware contents on the profile	The link from selection to conscious access in A1

Appendix C: Index of results

"Carried" means stated in the v1.1 working manuscript. "New" means not in that manuscript, not that it is original, and these results have had less review.

Here	Name	Basic move	Origin
Proposition 1	Garbling	Policy simulation	Carried (Prop. 1)
Proposition 2	Use without marginal information	Counting	Carried (Prop. 13)
Proposition 3	Shared-fate covariance	Total covariance	New
Proposition 4	Stationary condition	Lagrange	Carried (Prop. 2)
Proposition 5	Monotonicity	Feasible set	Carried (Prop. 3)
Proposition 6	Universal rationalizability	Lagrange and convexity	Carried (Prop. 14)
Proposition 7	Weighted water-filling	Lagrange and convexity	Carried (Prop. 4)
Corollary 1	Log-error conservation	Arithmetic	New
Corollary 2	Exclusion threshold	KKT conditions	New in this form
Proposition 8	Displacement	Counting	Carried (Prop. 5)
Proposition 9	Sharing versus serial access	Jensen, arithmetic	New; parts (b) and (c) restate known points
Proposition 10	Reader equivalence	Data processing	Carried (Prop. 6)
Proposition 11	Grouping invariance	Total-variation invariance	Carried (Prop. 15)
Proposition 12	Report information loss	Chain rule, data processing	Carried (Prop. 7)
Proposition 13	Introspective blind spots	Likelihood equality	Carried (Prop. 8)
Proposition 14	Expected progress	Bellman, Markov	Carried (Prop. 16)
Proposition 15	Restricted-protocol savings	Chain rule	Carried (Prop. 9)
Proposition 16	No advantage over joint erasure	Subadditivity	Carried (Prop. 10)
Corollary 3	No reason for a small summary	Subadditivity	Carried (§10.3)
Proposition 17	Tree-assisted erasure	Chain rule, spanning-tree exchange	Carried (Prop. 17)
Proposition 18	Observational equivalence	Likelihood equality	Carried (Prop. 11)
Proposition 19	Monitor anchor	Linear duality	Carried (Prop. 12)
Proposition 20	Oversight bound	Total-variation duality	Carried (Eq. 29)

Definitions 6–8 revise the v1.1 selection definition only in presentation. Hypothesis Q1 and the automorphism remark in Problem 45 are new.

Appendix D: Numerical checks

`v7_recheck.py` covers carried results and `paper_checks.py` covers new ones. Both use the standard library, assert every expected result, and passed on the final run. They are finite examples, not proofs, and they do not validate any architectural hypothesis.

Check	Result
Equation (36) on 1,000 random joints with deterministic summaries	Residual below 10^{-15}
Equation (14) against Blahut–Arimoto, three alphabet shapes	Solver never beats the target
All 16 one-bit functions in Proposition 2	Only XOR and XNOR give zero error
Equation (10) without side information	$D = 0.1100$ with four outputs; 0.25 for one-shot binary Z
Equation (37) against the rooted protocol, all trees of K_4, 100 joints	Gap below 10^{-15}
Table in §11.4	Asserted to three decimals
Equation (23) against numerical integration	Within 2×10^{-3}
Clock and dial, all same-macrostate pairs	4,004 pairs, 0 disagreements
Equation (32) against a forward law on 200 absorbing chains	Residual below 10^{-15}; drift never positive
Equation (33) and stationary average on 200 chains	Residuals below 10^{-14}
Corollary 1 on 500 random optimal allocations	Residual near 10^{-14}, which is close to an identity
Corollary 1 under allocation jitter and response noise	Products 0.098 and 0.282 at jitter 0.5 and 1; log statistic unchanged; noise shift 0.526
Set-size values in Problem 1	As stated
Corollary 2 by water-filling and by brute-force grid	Problems 4 and 6 as stated
Proposition 9, tolerance, switch-cost, retention, and completion cases	As stated
Equation (9) on 500 random parameter sets, and a global-state counterexample	Residual near 10^{-16}; counterexample gives 0.040
Equation (27) and recovered sensitivity	As stated
F1 and F2 example	As stated
Weak-cue accumulation	Total variation 0.05, about 0.38, above 0.8
Automorphisms	12 for the ring; 1 for irregular points on a line, verified to be a metric

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Entries marked † were cited from recall of the literature and were not confirmed by a search during preparation. Check them before relying on them. Unmarked entries were located by search or page fetch, and usually read at abstract level only.

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This work has not been peer-reviewed. Claude was used in this work, including in reviewing and helping execute tests. I hold the responsibility for it.